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3D printing of soft magnetic materials: From printing to applications

Feng-Hui Wang, Cai-Yin You *, Na Tian, He-Guang Liu, Jing Zhang, Xiao-Pei Zhu

School of Materials Science and Engineering, Xi'an University of Technology, Xi'an 710048, China

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ABSTRACT

Soft magnetic functional materials exhibit low coercivity, low power loss, and high permeability, making them highly utilized in power electronic devices. In some cases, soft magnetic components with intricate architectures are required to accomplish various tasks. The multi-material and multi-scale fabrication ability of three-dimensional printing (3DP) provides a new way for manufacturing soft magnetic devices with complex structures. The aim of the present work is to consolidate the latest advancements in the 3DP of soft magnetic materials (SMMs). The goal is to inspire future research directions by highlighting existing gaps in knowledge and potential areas for innovation. This paper summarizes recent research on 3DP of SMMs. Studies on 3DP of SMMs, such as Fe-Si and Fe-Ni alloys and soft magnetic ferrites and composites, are reviewed. Moreover, several mainstream 3DP procedures have been introduced. Furthermore, application scenarios of 3DP SMMs, including electronic components and magnetic-drive robots, are covered. Finally, the current challenges and future development trends of 3DP SMMs are discussed.

1. Introduction

Magnetic functional materials are used in various industries, including power electronics, communications, medicine, and aerospace, owing to their fast, long-distance, and accurate response in different environments [1]. Different magnetic materials and architectures can be used to create magnetic devices with various applications, including bionic soft robotics, sensors, shock absorbers, and magnetorheological dampers [2–4]. Magnetic materials are divided into hard [5], functional [6], and soft magnetic materials (SMMs) [7]. Because SMMs such as silicon steel [8], ferrites [9], permalloys [10], etc. exhibit low coercivity, low power loss, and high magnetic permeability [11–13], they are widely used in power equipment and electronic equipment. SMMs are utilized in the power sector at every step, from the production (generators) and transmission (transformers) of electric energy to its utilization through motors and reactors [14]. In the electronics industry, soft magnetic devices are utilized in the information exchange, transmission, and storage stages, including communication (inductors), automatic control, radio, television, power supply, electronic computing technology, and microwave technology that uses various ferromagnetic microwave devices [15].

The extensive use of SMMs in various fields has motivated the global materials community to focus on developing high-performance SMMs. The intricacy of soft magnetic devices required for various tasks makes their production challenging. Traditional manufacturing can no longer meet demands arising due to rapid industrial development. Traditional manufacturing can no longer meet the rapid development of industry and the rapid preparation of products with complex shapes and structures.

3D printing (3DP) [16,17], also referred to as additive manufacturing (AM), is a sophisticated manufacturing technology that directly forms parts from three-dimensional models [18,19]. AM is defined by the American Society of Testing and Materials (ASTM) as a process of joining materials to make objects from 3D model data, usually by layered stacking of the material, as opposed to subtractive manufacturing methodologies [20]. 3DP has the advantages of a high degree of freedom in design and manufacturing, short object forming cycles, gradient manufacturing of components and structures, personalization, and mass customization [21–23]. It combines advanced materials and digital manufacturing technologies and is an important part of the advanced manufacturing industry [24–26]. However, more in-depth research has been conducted on the 3DP structural materials than the 3DP soft magnetic functional materials. In recent years, several review articles on 3DP structural materials have been published [20,27–31]. Although some review articles on the 3DP of magnetic materials are available [32–36], there is no comprehensive and systematic review of research on the 3DP of soft magnetic functional materials.

This review discusses advances in the 3DP of SMMs. In Section 2, the

* Corresponding author.
E-mail address: caiyinyou@xaut.edu.cn (C.-Y. You).

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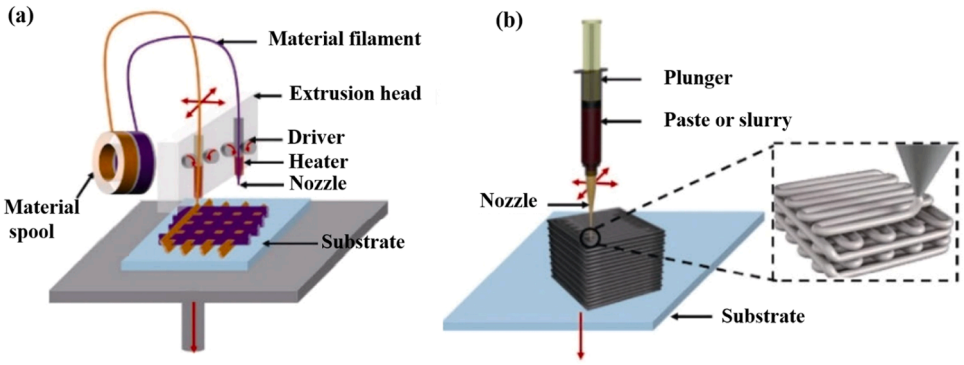

Fig. 1. Schematic diagram of material extrusion process [54] performed using (a) FDM and (b) DIW methods.

research progress in this field is reviewed. It covers 3DP techniques such as material extrusion (ME), powder bed fusion (PBF), binder jetting (BJ), and inkjet printing (IJP), which are commonly used for manufacturing SMMs. Furthermore, studies on 3DP of FeSi alloys, FeNi alloys, soft magnetic ferrites, and soft magnetic composites are discussed regarding raw material preparation, process optimization, post-treatment, and soft magnetic properties. Section 3 details the application of 3DP of SMMs. Finally, Section 4 discusses the challenges and future development of 3DP of SMMs.

2. Research on 3DP of SMMs

The earliest research on 3DP of SMMs was conducted on pure iron by adopting the PBF technology. Further developments in 3DP technologies enabled the preparation of other SMMs. First, this section summarizes the 3DP technologies commonly used for manufacturing SMMs. Next, the studies on 3DP of FeSi and FeNi alloys, soft magnetic ferrites, and soft magnetic composites are reviewed.

2.1. 3DP technologies for manufacturing SMMs

Different types of 3DP can be categorized based on the kind of material. The International Organization for Standardization (ISO) divides it into seven general types: photopolymerization (VP) [37], ME [38], PBF [24], material jetting (MJ), BJ [39], direct energy deposition (DED) [40,41], and sheet lamination (SL) [42]. The 3DP technologies for manufacturing SMMs mainly include ME, PBF, BJ, and IJP [43].

2.1.1. Material extrusion

Material extrusion is a commonly used technology for fabricating compatible materials. In this process, materials are extruded through nozzles. In particular, a material filament is melted using a heated nozzle, and the molten material is deposited on the platform by a printer along the path determined by the slicing software. Then, the molten material cools down and solidifies to form a sample [44–47]. Fused deposition modeling (FDM), one of the earliest commercial 3DP technologies, is the most mature and popular ME technology. It is shown in Fig. 1a. Similar to all 3DP technologies, FDM starts by preparing a digital model, which is then converted into instructions to be followed by a 3D printer. The filament on the FDM spool is loaded into the 3D printer, which feeds it to the printer nozzle in the extrusion port. The nozzle is heated to the desired temperature, causing the filament to melt, and the continuously stacked layers combine as they solidify to form a component [48]. The magnetic properties of soft magnetic parts formed by FDM are affected by the orientation and concentration of magnetic particles [49]. The proportion of magnetic particles must be higher than 65% to ensure satisfactory magnetic properties [50].

Another ME approach is direct ink writing (DIW, Fig. 1b) [51]. DIW is commonly employed in producing 3DP magnetic devices because of its superior polymer selection tolerability [52]. Due to their strong multi-material printing capabilities, DIW SMMs can be used in bone regeneration and magnetic robots. It should be noted that low-viscosity liquid polymers and nano-magnetic particles must be chosen as the raw materials to prevent the print port from being blocked [53].

2.1.2. Powder bed fusion

PBF is a 3DP technology that uses energy sources such as laser or electron beam to fuse granular materials such as metal, ceramic, or polymer to form 3D objects. Selective laser melting [55,56] (SLM) and selective electron beam melting [57] (SEBM) are common PBF technologies.

SLM is a sophisticated manufacturing technique used to produce alloy parts with intricate structures and forms. It employs a laser as its energy source that melts powder layer by layer, selectively, along a

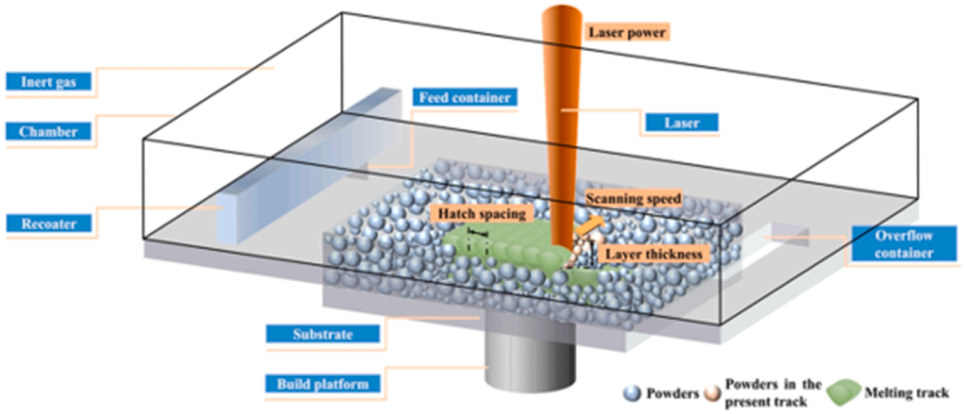

Fig. 2. Schematic diagram of the SLM principle [58].

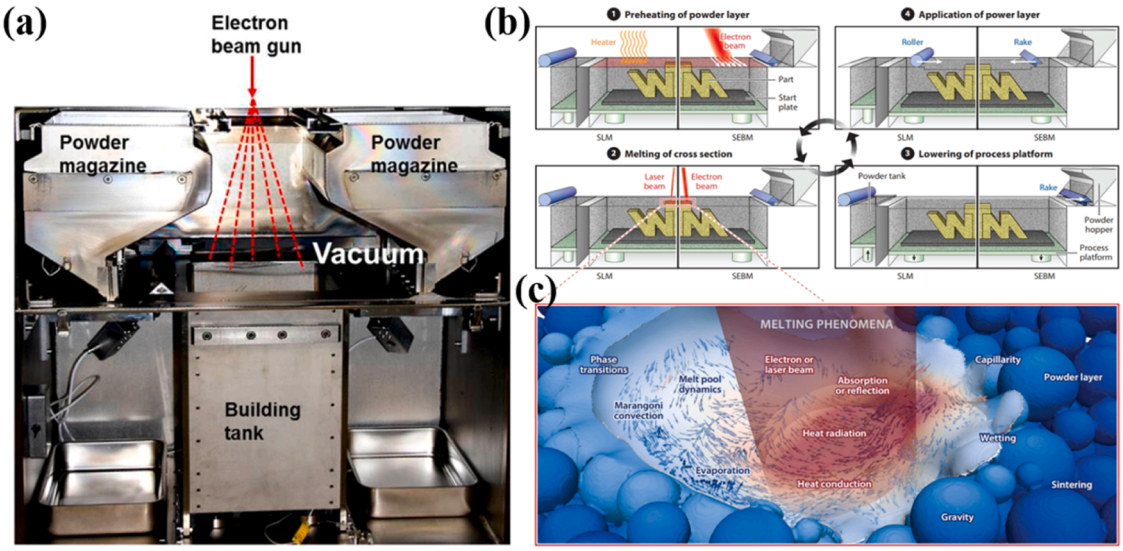

Fig. 3. SEBM technology (a) printing room of the equipment [71], schematic diagram [64] of (b) printing process, and (c) physical phenomena.

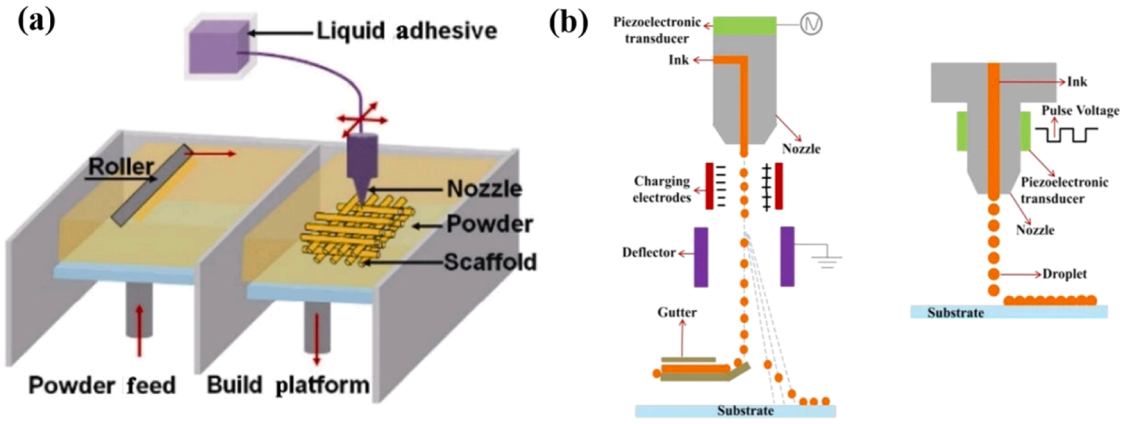

Fig. 4. Schematic diagram of (a)BJ [54] and (b)IJP [76].

predetermined path and achieves nearly net-shaped components. The high solidification rate of SLM (10^5-10^6 K/s) results from its rapid melting and solidification process, which also helps produce materials with refined grain size due to supercooling, thus enhancing the mechanical properties of components. Fig. 2 shows a schematic diagram of the SLM principle.

On the other hand, in SEBM, a metallurgical process that rapidly melts and solidifies micro-molten pools, high-energy electron beams are utilized to target metal powders with a particle size distribution of 45–106 μm [59–62]. The high vacuum and spontaneous heat treatment cycle [63] in the SEBM process set it apart from SLM and cause the specimen's residual stress to be lower than that of the SLM specimen.

Fig. 3 shows the printing room of SEBM equipment, the SEBM printing process, and its underlying physical phenomena. The printing process is divided into four steps: powder laying, preheating, melting and solidification, and platform falling. The above four steps are repeated until the last layer of printing, thus completing the whole process [64]. In the PBF process, melting and solidification of powder, Marangoni convection in the molten pool, and phase change occur on macroscopic, mesoscopic, and microscopic scales, respectively. Furthermore, because thermal convection, thermal radiation, and heat conduction also occur, PBF is a complex thermodynamic and kinetic process.

Numerous studies have been published on SLM-printed soft magnetic alloys [65–70]. These studies examined the impact of post-processing on microstructures and soft magnetic properties in addition to the effects of laser process parameters, including laser power and scanning speed, on magnetic properties. To our knowledge, no reports on SEBM printing of SMMs exist.

2.1.3. *Binder jetting*

BJ (Fig. 4a) is a 3DP technology that forms parts by spraying binders. BJ is mainly used to print polymer materials, metals, and ceramic materials. Some scholars have tried this method to print SMMs [72,73]. Diversification of soft magnetic equipment would enable BJ methods to prepare magnetic components with complex structures by using powder as a support structure. Thus, BJ shows more application prospects in manufacturing new soft magnetic components.

2.1.4. *Inkjet printing*

IJP(Fig. 4b) is a non-contact micron-scale printing process that can be achieved by directly spraying a nano-sized solution on a flexible or hard substrate. IJP is a promising printing process since it can directly form patterned films without masks. IJP is widely used for manufacturing thin-walled soft magnetic substrates owing to its high resolution and multi-material printing capabilities [74]. IJP requires an appropriate selection of low-viscosity liquid polymer and nanoscale magnetic particles as the matrix binder and functional fillers,

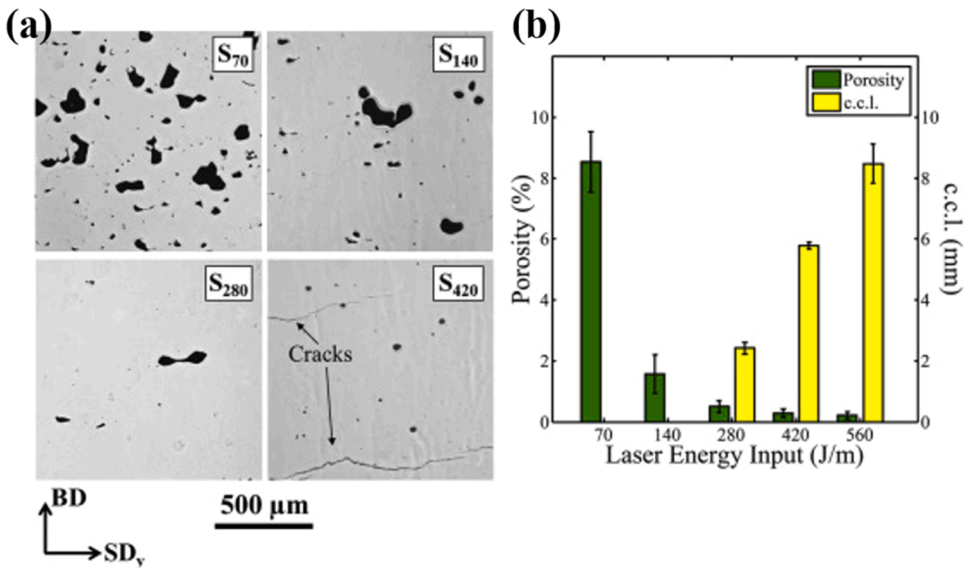

Fig. 5. Effect of laser energy density on porosity and crack formation of Fe-6.9 wt%Si samples [77] (a) Optical micrographs of different samples (b) percentage porosity and cumulative crack length (c.c.l.) versus laser energy input.

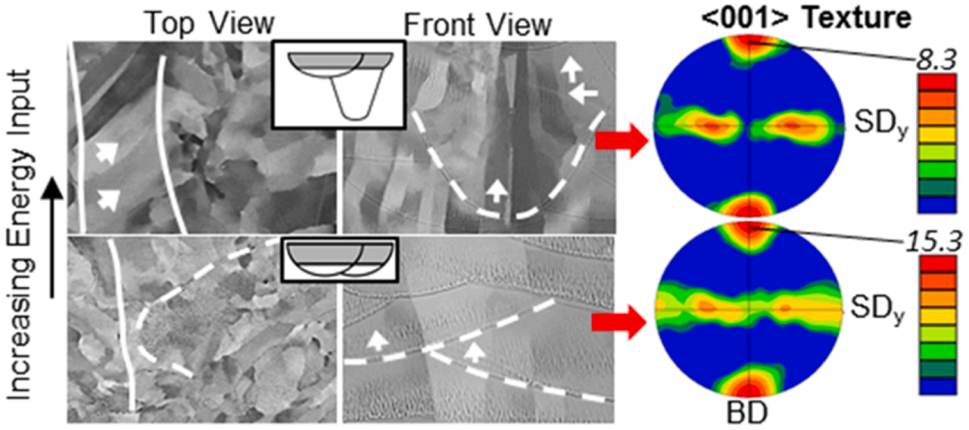

Fig. 6. Cross-sectional morphology of molten pool and pole figure under different energy inputs [77].

respectively, to avoid nozzle clogging [75].

2.2. Research on 3DP of SMMs

SMMs are indispensable in electrical equipment such as transformers, inductors, and motors. Furthermore, it is possible to enhance the magnetic characteristics of soft magnetic components through intricate structures built using 3DP, which offers exceptional design and manufacturing flexibility. Research on 3DP of SMMs such as soft magnetic alloys, ferrites, and composites is mainly focused on the potential relationship between processing, post-treatment, microstructure, and material properties. This subsection discusses the research progress in 3DP of various SMMs.

2.2.1. Fe-Si alloys

Fe-Si alloys, also known as silicon steel, are formed by adding a small quantity of silicon to pure iron. The saturation magnetic (M_s) induction of Fe-Si alloys can reach 2 T. Owing to their good magnetoelectric properties, easy mass production, and low cost, they have been widely used in power electronic equipment, including transformers, inductors, sensors, and motors. In traditional manufacturing processes, Fe-Si alloys are first manufactured as laminated sheets that are stacked to form a magnetic core. 3DP provides a new idea for forming Fe-Si alloys.

Laser 3DP technology is a favorable tool for manufacturing Fe-Si alloys. To promote the densification of parts, Garibaldi [77] et al. studied the effect of laser energy density (LED, changing scanning speed) on the densification of Fe-6.9 wt%Si. They also studied the obtained microstructures in terms of solid phase formation, grain size, morphology, and texture. As shown in Fig. 5, the as-built SLM sample has irregularly shaped macropores with laser energy input E<280 J/m. At E=280 J/m, crack formation is initiated, although the irregular pores are reduced. When E≥420 J/m, small bubble-like voids (i.e., spherical holes) appeared, further promoting crack formation.

Furthermore, in [77], the columnar grains of the as-built samples were observed to grow along the building direction (BD). This finding is consistent with Goll's study [78]. As the laser energy is increased, the <001> fiber texture changes to cube texture along the BD, as shown in Fig. 6 [77]. Moreover, the grain size of the as-built SLM-ed Fe-6.9 wt%Si sample is in the range of 10–20 μm. Another study reported the optimal grain size in terms of minimum energy loss to be 150 μm [79]. As-built SLM Fe-6.9 wt%Si sample was annealed at 1150 °C for 1 h to produce a recrystallized microstructure consisting of equiaxed grains with sizes up to 300 μm (Fig. 7) [80]. Furthermore, as shown in Fig. 8, the magnetic properties of SLM Fe-6.9 wt%Si samples change with annealing temperature. The approximate values of maximum permeability (μ_{max}) of the as-built and annealed samples are 2000 and 24000, respectively

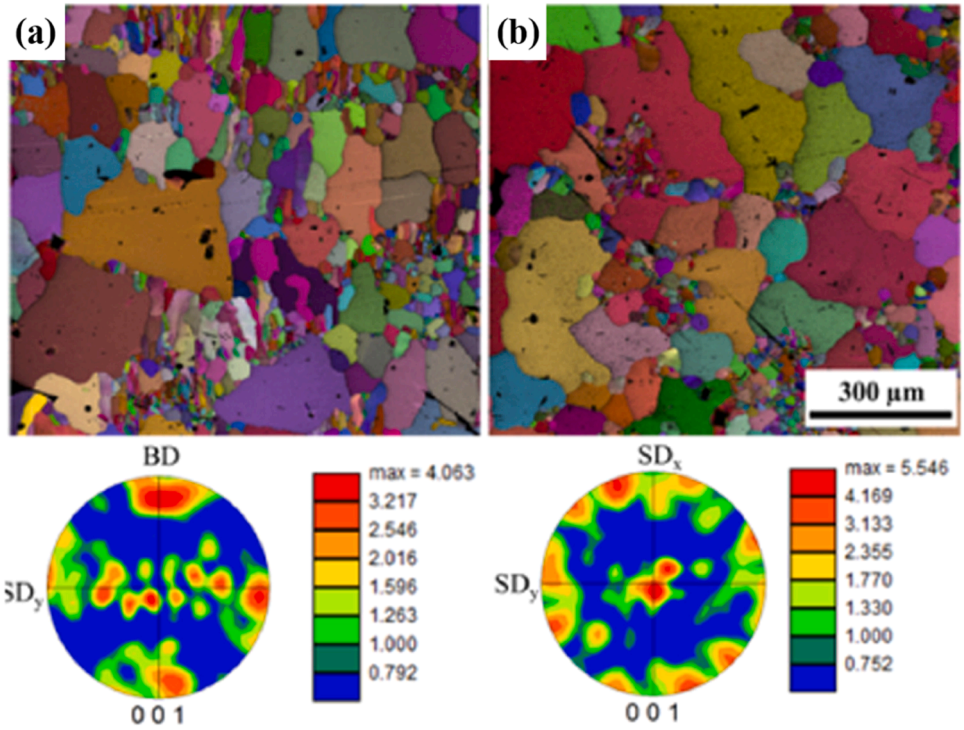

Fig. 7. Electron Backscatter Diffraction (EBSD) images of the SLMed Fe-6.9 wt%Si samples annealed at 1150 °C for 1 h [80]: (a) front views (b) top views.

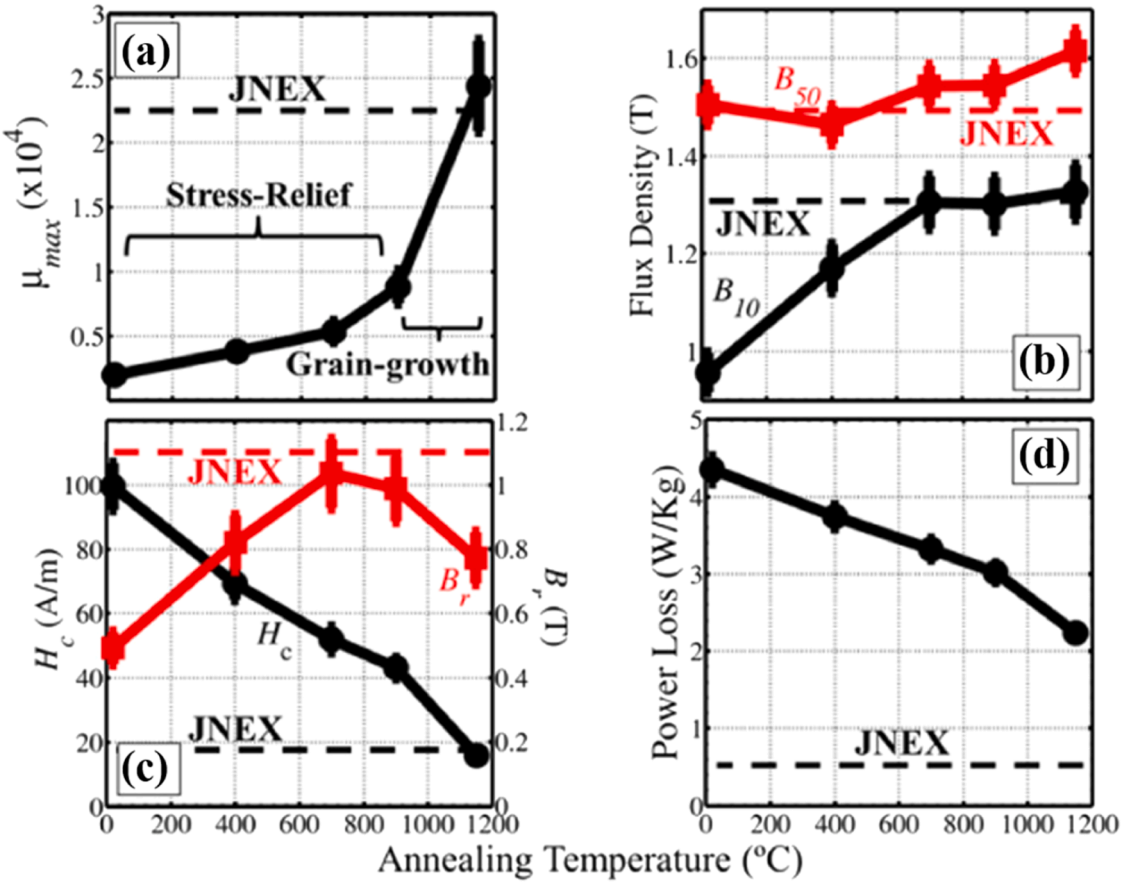

Fig. 8. Magnetic properties of SLM Fe-6.9 wt%Si samples with annealing temperature [80]: (a) μ_{max} (b) B_{10} and B_{50} (c) H_c and B_r (d) power loss at $B_{max} = 1$ T and $f = 50$ Hz.

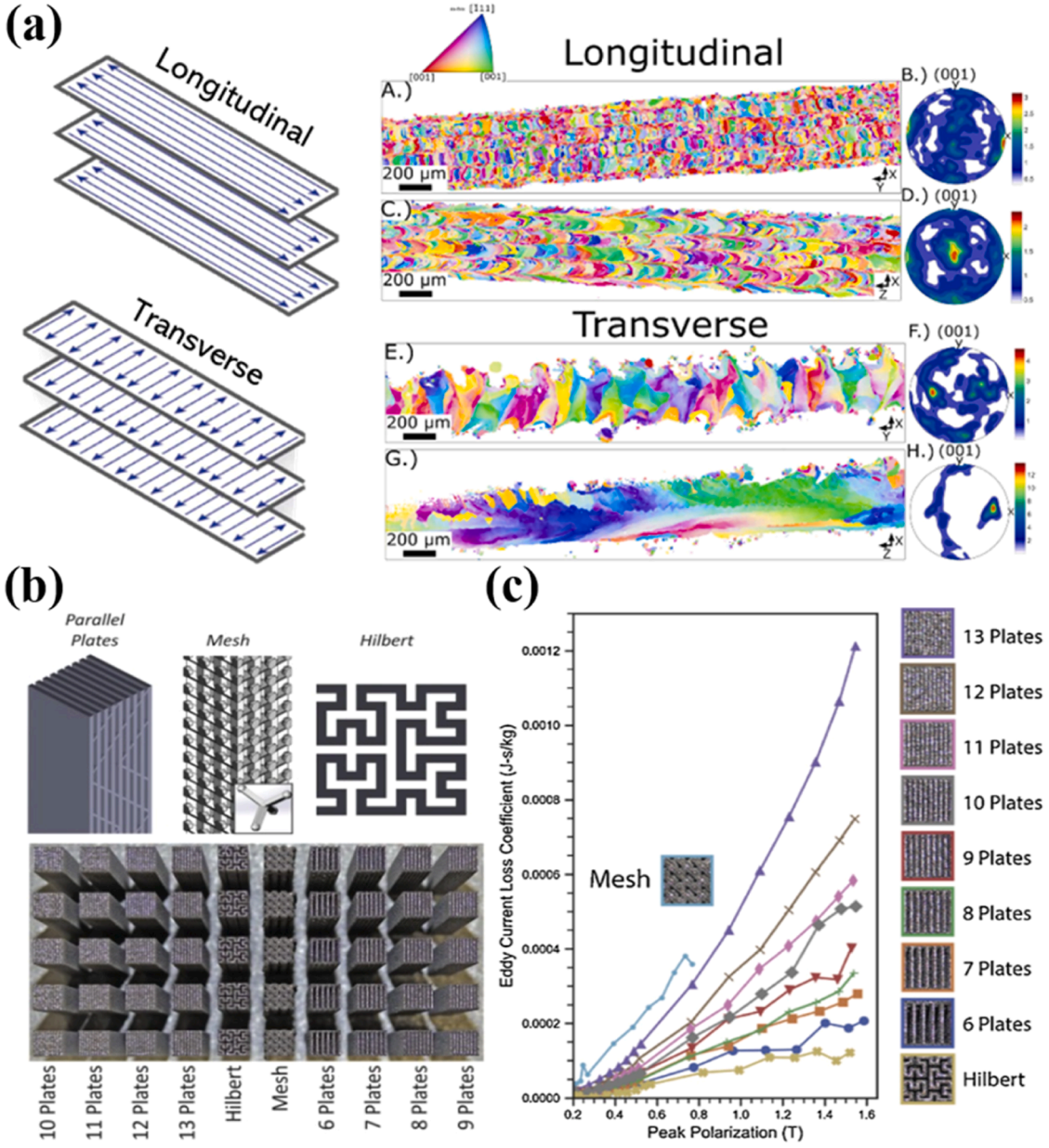

Fig. 9. Effect of scanning mode and geometry on the microstructure and eddy current loss coefficient of Fe-Si alloy-based samples obtained using SLM [84].

[80], and high-temperature annealing leads to a minimum coercivity (H_c) of 16 A/m. Garibaldi et al. studied the relationship among laser energy input, microstructures, and magnetic properties of SLMed Fe-6.9 wt%Si [81]. They reported that, under a high LED, a cube texture is formed, which reduces the soft magnetic properties. Meanwhile, samples with density>99 % and strong crystallographic texture along BD showed optimal magnetic properties. Hence, they concluded that porosity and texture are the main factors that affect magnetic properties.

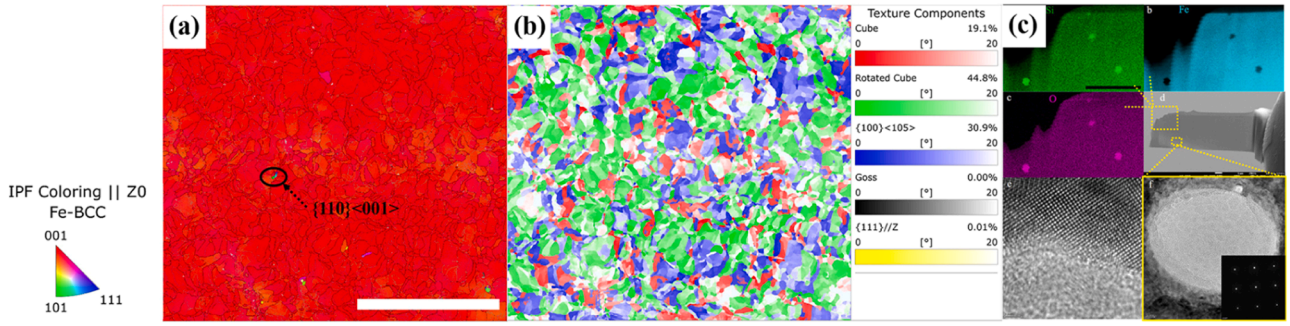

Fig. 10. (a) inverse pole figure, (b) texture figure, and (c) amorphous precipitates analyzed by TEM and SEM-EDS of L-PBF Fe-3.5 wt%Si alloy [93].

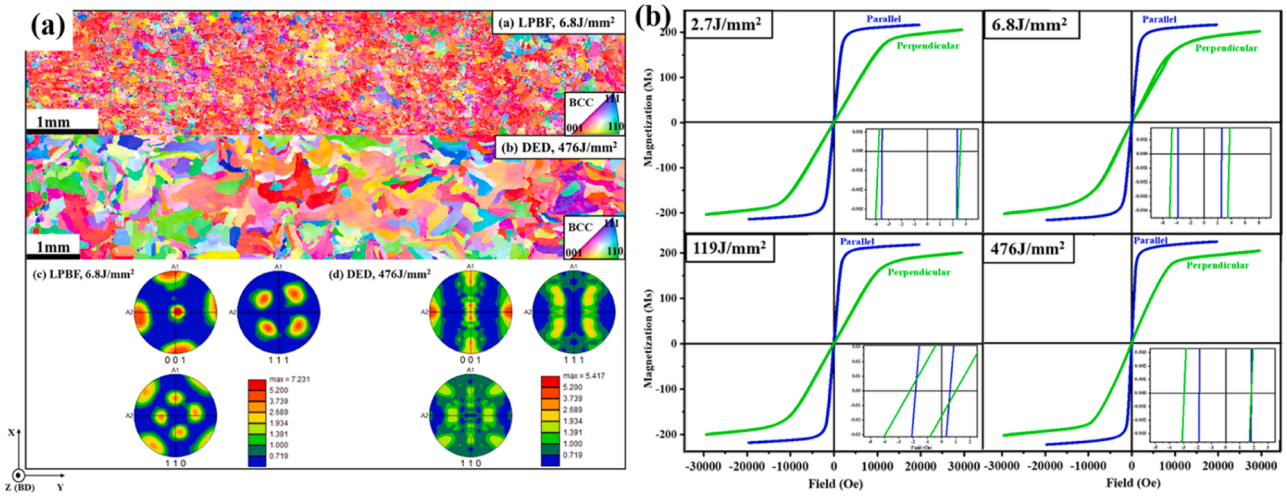

Fig. 11. L-PBF and DED Fe-3.8 wt%Si alloys [94]: (a) IPF maps, inverse pole figures, and pole figures; (b) hysteresis curves.

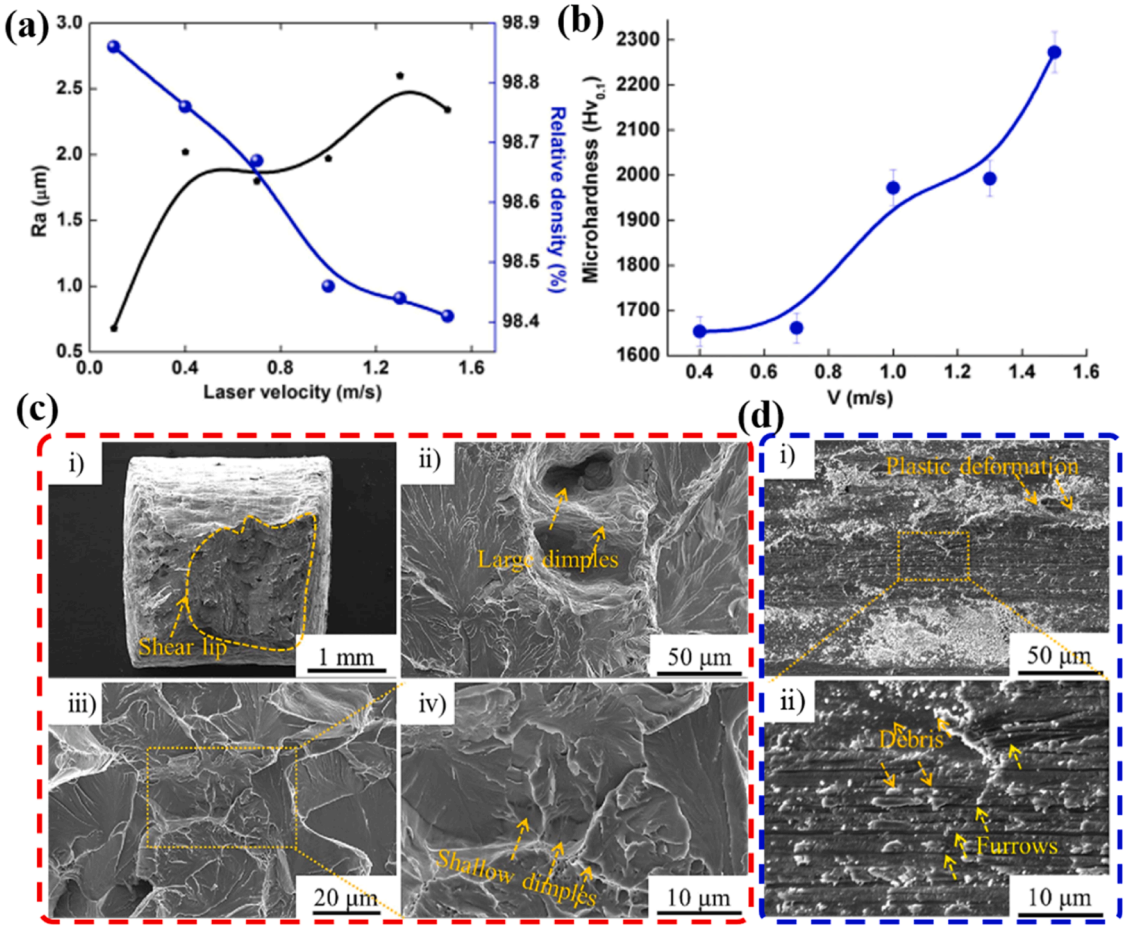

Fig. 12. Effect of laser velocity on (a) R_a and ρ_r (b) microhardness of SLM Fe-Si alloys [95] and SEM image of SLM Fe-3 wt%Si [96], (c) tension fracture, and (d) friction and wear surfaces.

The microstructure of as-built SLM-ed Fe-Si alloy parts was improved by heat treatment [80,82–84]. In [82], the as-built sample was annealed at 1000 °C for 3 h, causing the grain size to increase significantly, thus eliminating the stress. Therefore, a combination of quenching and annealing processes is expected to yield a more uniform structure with improved soft magnetic properties. Other studies [85–89] have also explored the process parameters and heat treatment while aiming to clarify the potential relationship between these factors and microstructure and soft magnetic properties to provide a theoretical basis for practical applications.

The scan pattern and geometry also affect the microstructure and soft magnetic properties of 3DP Fe-Si alloy samples [84,90–92]. Plotkowski et al. studied the effects of scanning mode and geometry on the microstructure and soft magnetic properties of Fe-Si alloys fabricated using

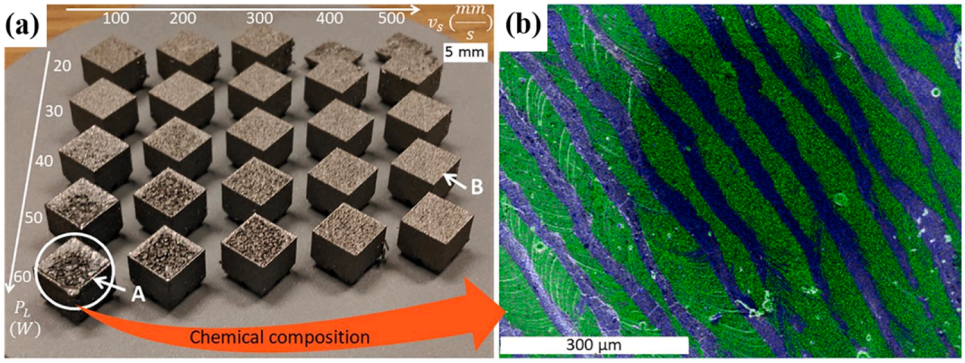

Fig. 13. Permalloy parts obtained using LBM [113]:(a)Samples under different parameters (b) Surface element distribution of sample A in (a) at a volume energy density of 428 J/mm^3.

AM [84]. EBSD (Fig. 9a) results show that a longitudinal scan pattern produces more misoriented and finer grains, whereas the transverse scan pattern yields higher epitaxial growth in the BD. The several complex cross-sections manufactured for magnetic testing are shown in Fig. 9b. Their eddy current loss coefficients are plotted in Fig. 9c, which shows that the influence of geometry is significant. Increasing the number of thin walls in the parallel plate samples increases the eddy current loss coefficient. Moreover, the mesh structure shows poor performance relative to the other cross-sections. The Hilbert curve shows a significant improvement over the parallel plates and the best performance of the tested samples [90].

The texture and microstructure of Fe-3.5 wt%Si alloy via laser powder bed fusion(L-PBF) have been studied (Fig. 10) [93]. Strong λ-fiber textures including {001}<001> Cube texture, {001}<110> rotated cube texture, and {100}<105> texture were obtained on the top surface of as-built Fe-3.5 wt%Si sample [93]. Amorphous precipitates were found for the first time in as-built LPBF Fe-3.5 wt%Si parts, as shown in Fig. 10c [93]. To clarify the effect of λ-fiber texture on the magnetic properties of LPBF Fe-Si alloys, Varahabhatla [94] studied Fe-3.8 wt%Si alloys using L-PBF and DED techniques. As shown in Fig. 11a, compared with the L-PBF samples, the DED samples had rough grains and random texture and did not show Cube and λ-fiber textures [94]. All samples exhibit good soft magnetic properties(Fig. 11b), with M_s values ranging from 205 to 232 emu/g and H_c values ranging from 1.2 to 4.2 Oe [94].

In addition to the soft magnetic properties, the mechanical properties of AM Fe-Si alloys have also been studied. SLM Fe-Si samples exhibited high density, low surface roughness, and high microhardness (1654–2273HV), as shown in Fig. 12(a-b) [95]. Gao reported a microhardness of 235 $HV_{0.1}$ and an elongation of 8.8 % for the as-built SLM Fe-3 wt%Si magnet [96]. Moreover, its ultimate tensile strength and yield strength are reported to be 562 MPa and 445 MPa, respectively. A mixed mode of ductile and brittle failure is the main rupture mechanism for the SLM Fe-3 wt%Si magnet [96], as shown in Fig. 12**c**. The SLM Fe-3 wt%Si magnet exhibits good wear resistance with a wear rate of 5.3×10^{-5} mm^3/(N·m). Furthermore, as shown in Fig. 12**d**, the dominant wear mechanisms are adhesive and abrasive wear [96]. Fully dense soft magnetic Fe-Si stators have been successfully printed using BJ [73,97]. The obtained 1.02 mm-thick samples had nearly 99 % dense parts without cracks, an ultimate tensile strength of 434 MPa, H_c = 32 A/m, M_s = 1.83 T, and a maximum relative permeability of 10500 [73].

2.2.2. Fe-Ni alloys

Fe-Ni alloys are used in magnetic shielding, magnetic cooling, magnetic sensors, and power transformers owing to their high permeability, low coercivity, low magnetostriction, and anisotropic magnetoresistance [98–101]. The research on 3DP of FeNi alloys has gained considerable attention from researchers [102–107]. Zhang et al. reported that the laser energy density strongly affects the soft magnetic properties of L-PBF Fe-Ni alloys [108]. They achieved a low bcc lattice parameter and large grain size at a laser scanning speed of 0.4 m/s and a laser power input of 110 W, with H_c = 88 A/m and M_s = 565 emu/g. SLM Fe-80 %Ni alloys have low coercivity and high saturation magnetization [109], which is consistent with the behavior of Fe-Ni alloy materials produced by casting or sintering processes [110,111]. Shishkovsky et al. studied the optimal process of SLM Fe-Ni alloys to analyze the effect of external magnetic field (EMF) on the growth structure during solidification and preparation of high-density samples [112]. The optimal energy density range for preparing Fe-Ni alloys by SLM is 60–100 J/mm. Furthermore, the EMF affects the phase redistribution under the SLM process and aggravates iron oxidation [112]. Schönrath simultaneously prepared 25 permalloy parts with different energy densities via Laser Beam Melting (LBM), as shown in Fig. 13 [113]. The volume energy density was 438 J/mm^3, and a stripe-like segregation of iron and nickel was observed at the top layer, resulting in a large

Table 1
Research on laser additive manufacturing of FeNi alloy-based materials and the influence of process conditions on phase structure and soft magnetic properties [35].

Processing condition	Phase(s)	M_s(emu/g)	H_c(Oe)	Ref.
Fe-80Ni				
P=100 W,v=0.1 m/s	$FeNi_3$	99	~36	[109]
P=100 W,v=0.2 m/s	$FeNi_3$	~98	~35	
P=100 W,v=0.3 m/s	$FeNi_3$	~96	~36	
P=100 W,v=0.4 m/s	$Fe_{70}Ni_{30}$+Ni	95	~37	
Fe-30Ni				
P=300 W,v=5 IPM	Majority of FCC	120	~30	[105]
P=300 W,v=20 IPM	Majority of BCC	165	~25.87	
P=500 W,v=5 IPM	FCC	125	~19	
P=500 W,v=20 IPM	Majority of BCC	152	~21	
P=300 W,v=8 IPM	Majority of FCC	95	~47	[114]
P=400 W,v=8 IPM	Majority of FCC	99	~13	
P=500 W,v=8 IPM	FCC	105	~10	
P=400 W,v=20 IPM	Majority of FCC	103	~32	
P=300 W,v=13 IPM	FCC+BCC	96	~44	
Ni-21.5Fe				
E=428.57 J/mm^3	$FeNi_3$	97		[115]
E=71.43 J/mm^3	$FeNi_3$	94.8		
Ni-15Fe-5 V				
P=400 W,v=20 IPM	FCC	72	~6	[104]
Ni-15Fe-5Mo				
P=400 W,v=20 IPM	FCC	82	~4.9	[104]
P=200 W,v=0.4 m/s+ annealing	FCC	78	~1.88	[116]

Table 2
M_s and H_c of $Ni_{100-x}Fe_x$ alloys in different systems.

Technique	Composition	M_s(emu/g)	H_c(Oe)	Ref.
Laser additive manufacturing (LENS)	$Fe_{30}Ni_{70}$	119.8	5.5	
	$Fe_{40}Ni_{60}$	129	4.5	
	$Fe_{55}Ni_{45}$	161.6	4.9	[117]
	$Fe_{60}Ni_{40}$	165.6	6.2	
	$Fe_{70}Ni_{30}$	168.7	7.9	

uniaxial magnetic anisotropy and uniform distribution of internal elements. Table 1 summarizes some studies on laser additive manufacturing for Fe-Ni alloys [35].

Varun Chaudhary [117] was the first to use Laser Engineering Net Shaping (LENS) to fabricate the functionally graded SMMs $Ni_{100-x}Fe_x$. LENS $Ni_{100-x}Fe_x$ alloys exhibit a mixture of fcc and bcc phases [117]. As the Fe concentration increases, M_s increases, whereas H_c decreases initially and then increases. Table 2 gives the M_s and H_c under different systems. the as-built LENS $Ni_{100-x}Fe_x$ series have a hardness of approximately 160 VHN [117]. Some scholars have tried to manufacture Fe-Ni alloys using FDM. Arbaoui [118] et al. prepared a soft magnetic composite material composed of $Ni_{81}Fe_{19}$ and Polyethylene (PE) and prepared the filament with a double-screw extruder (Fig. 14a) to study the feasibility of forming NiFe/PE composites by FDM technology (Fig. 14b). A magnetic part (Fig. 14c) was successfully prepared, indicating the feasibility; however, no performance test was conducted. In [119], Ni-based electrode materials were prepared for the first time using digital light processing (DLP) technology. The reported electrochemical analysis results show that the Ni-Fe electrode requires a higher overpotential to reach a current density of 50 mA/cm.

Patton et al. mixed polylactic acid (PLA) polymer with 40 wt% iron powder to prepare magnetic filaments for fused filament fabrication (FFF) [120]. They pointed out that when the EMF is perpendicular to the building direction, magnetic susceptibility increases as the amount of magnetic particle filler is increased. A highly loaded suspension of iron oxide nanoparticles (IOPs) was developed for DIW 3DP, as shown in Fig. 15a, with Fe_2O_3 nanoparticles in these suspensions reaching 81 wt% [121]. The results indicate that the ink and as-built DIW IOPs samples

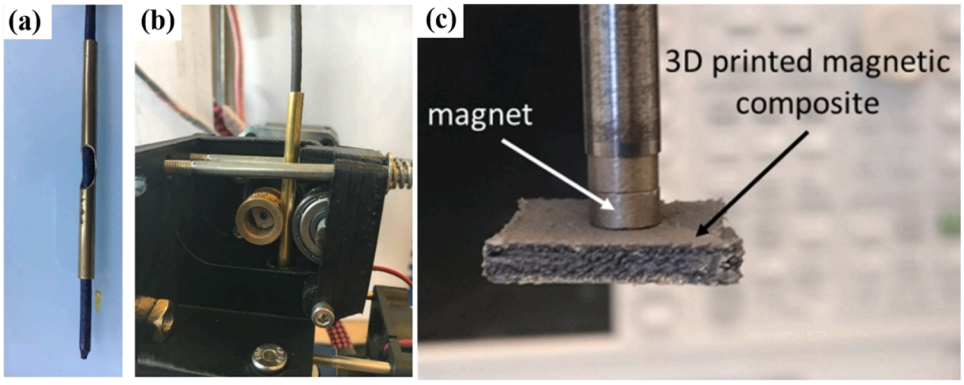

Fig. 14. FDM-based formation of NiFe/PE composite [118] (a) wire, (b) FDM equipment, and (c) as-built sample.

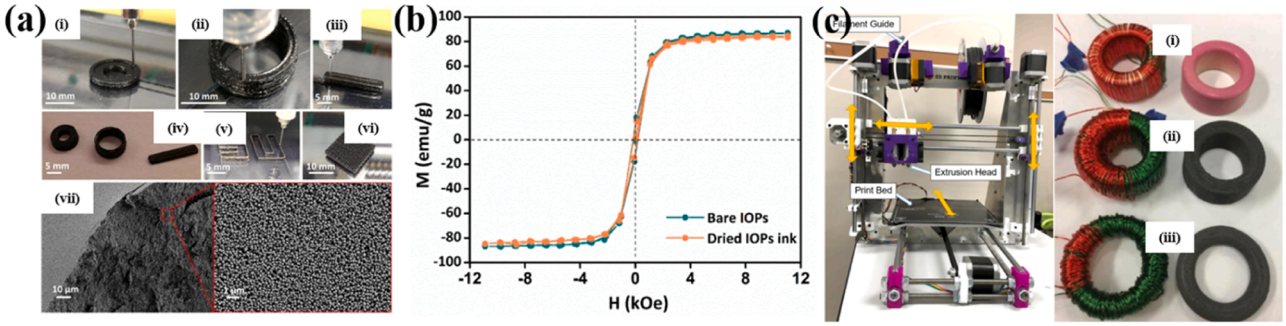

Fig. 15. (a) 3DP IOPs [121], (b) soft magnetic properties of ink and IOP samples [121], and (c) FDM printed ring transformer core [122].

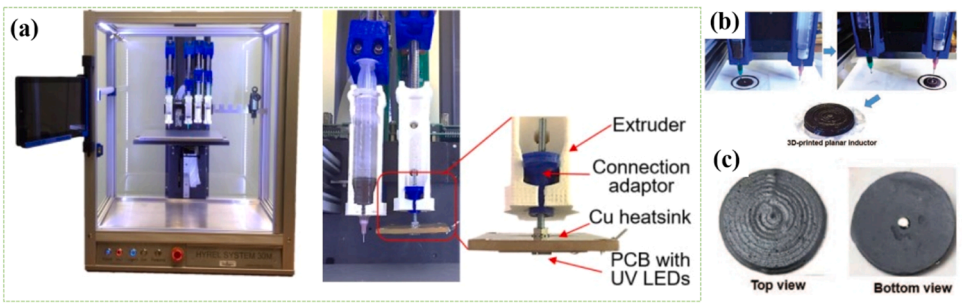

Fig. 16. UV-assisted extrusion 3DP ferrites: (a) extrusion 3DP equipment and UV-assisted devices [130]; (b) 3DP ferrite inductors; and (c) sintered samples [129].

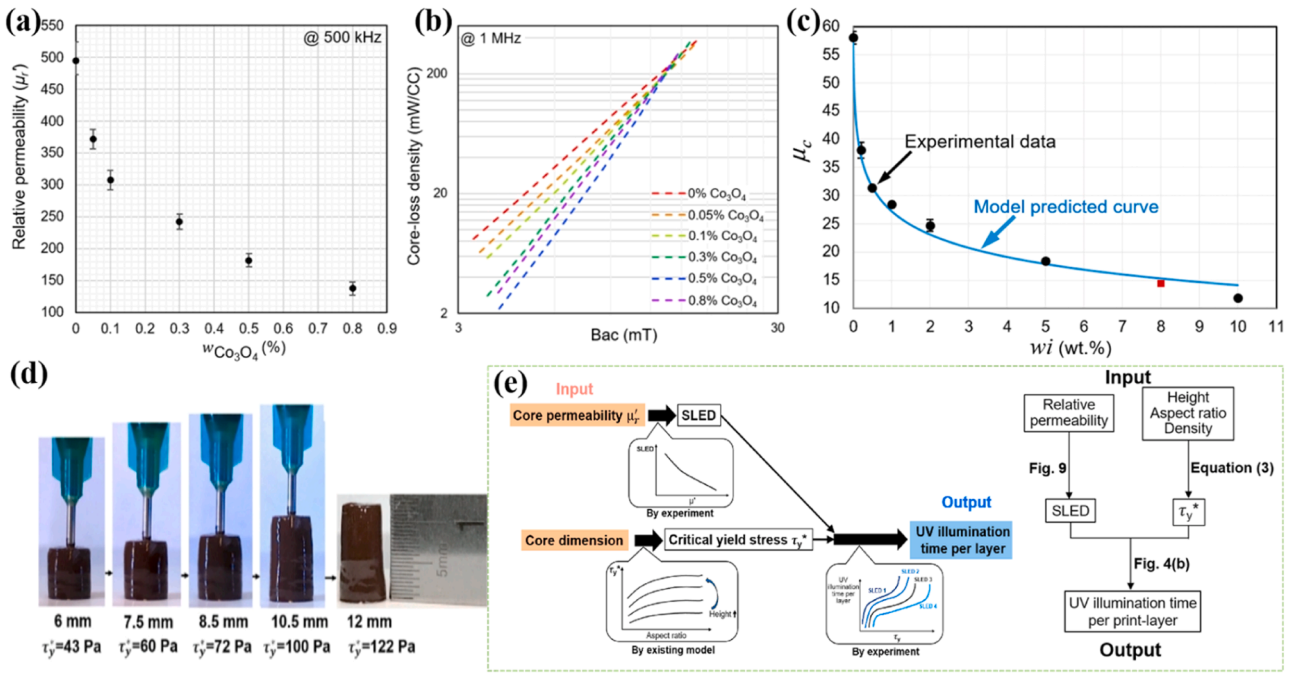

Fig. 17. (a) Relative permeability and (b) core-loss density of 3DP NiCuZn soft magnetic ferrites with different Co_3O_4 contents [131]. (c) Comparison of experimental data and model prediction [132], (d) sample heights, and (e) guide of 3DP assisted by UV curing [128].

have similar soft magnetic properties, as shown in Fig. 15b. Bollig et al. prepared a ring transformer core using FDM, as shown in Fig. 15c [122]. The filament is a mixture of PLA polymer with 40 wt% iron particles.

2.2.3. *Soft magnetic ferrites*

Soft magnetic ferrites exhibit high permeability and resistivity and low loss at high frequency. These are widely used in communication, sensing, audio, and video equipment, switching power supply, and magnetic head industry. It is a core material of electronic components. Some studies have been reported on 3DP of soft magnetic ferrites [123–128]. Most of these have focused on Mn and MnZn ferrites (MZF) and Ni and NiZi ferrites (NZF). In different studies, Lu et al. developed a UV-curable sintered NiCuZn [129] and NiZn [130] ferrite slurry and prepared ferrite cores by UV-assisted extrusion 3DP, as shown in Fig. 16. By changing the slurry composition and sintering temperature, some desirable properties have been achieved: 1) NiZn ferrite core with a maximum relative permeability of 103 and a resonance frequency of 30 MHz [130] and 2) NiCuZn ferrite inductors with inductance and resistance of 792 nH and 15 mΩ, respectively [129].

Based on [129], Liu et al. added Co_3O_4 to the NiCuZn ferrite (NCZF) slurry to explore the applicability of Co_3O_4 in tailoring the permeability of the 3DP NCZF cores [131]. The results showed that, as the weight fraction of Co_3O_4 oxide increases from 0 % to 0.8 %, the relative permeability of 3DP NCZF decreases from 459 to 138, as shown in Fig. 17a. The core-loss density of NCZF decreased by 77.6 % with 0.5 % Co_3O_4 addition at low magnetic flux density [131], as shown in Fig. 17b. Liu pointed out that the effect of Co_3O_4 on the permeability and core-loss density of ferrites is due to the pinned domain wall motion and the reduced electron rushing from Fe^{2+} to Fe^{3+} in the ferrite [131]. Furthermore, Liu [132,133] studied the effect of SiO_2 addition on the relative permeability of NCZF and established a mathematical model to accurately predict the resultant change in the relative permeability of the soft magnetic ferrite core (Fig. 17c) [132]. Liu [128] reported that sample slumping initiated when the printing height of the NCZF sample was 12 mm, as shown in Fig. 17d. Because yield stress is affected by the UV curing time during the 3DP process, a guide rule (Fig. 17e) was developed based on some slumping models, and the shortest curing time for each printing layer was specified to prevent the sample from collapsing.

Ding et al. studied 3DP NiZn soft magnetic ferrite based on DIW. In [134], NiZn ferrite samples with multi-layer mesh architecture were prepared using DIW accompanied by heat treatment, as shown in Fig. 18a. SEM analysis results showed that the network NZF samples had well-sintered polycrystalline micro-sized grains without pores [134]. These are shown in Fig. 18b. The hysteresis loop (Fig. 18c) shows that the M_s is approximately 77.5 emu/g at room temperature, and the H_c is negligible [134]. Wei [134] pointed out that these mesh architecture ferrites are expected to be used as magnetic filters to capture magnetic nanoparticles effectively. Peng [135] altered the type of metal oxides and successfully prepared bulk Ni ferrite (Ni-Fe_2O_4) using FFF 3DP (Fig. 18d). After sintering, the sample exhibited dense polycrystalline microstructure with satisfactorily low porosity (Fig. 18e), saturation magnetization of 48.39 emu/g, and relative density of 93 % [135]. Moreover, Ni ferrite samples with unique geometric shapes were printed using this method, as shown in Fig. 18f. Based on [134], Wei [136] et al. prepared a slurry by changing the Zn concentration and fabricated a network-structured NZF sample using the same approach(Fig. 18g). Fig. 18h shows that the M_s of NZF reaches a maximum value of 77 emu/g when the zinc concentration is 0.4 [136]. This can be attributed to the cationic distribution and the grain growth with extraordinarily high crystallinity [136]. Similarly, An et al. used DIW technology to print a NiZn soft magnetic ferrite and studied the basic relationship between the rheological properties of NZF slurry and its DIW printability [137]. Furthermore, they successfully printed a transformer with an EFD core, as shown in Fig. 18i.

FDM has also been applied for printing soft magnetic ferrites. Wang [138] et al. mixed the melted acrylonitrile-butadiene-styrene(ABS) with NiZn ferrite powder to prepare a filament (Fig. 19a) and used it to print a NiZn ferrite sample with a diamond lattice structure (Fig. 19b) using FDM. Aqzna doped a small amount of Zn ferrite in ABS to study the effect of varying Zn ferrite content on the mechanical and electrical properties of FDM ABS/$ZnFe_2O_4$ composites, as shown in Fig. 19c [139,140].

IJP is considered to be a powerful tool for manufacturing soft magnetic ferrites. Kang et al. prepared ferrite ink for IJP by mixing NiZn soft magnetic ferrite powder with insulator ink to print a flat spiral inductor with soft ferrite film on a polyimide (PI) substrate, as shown in Fig. 20a

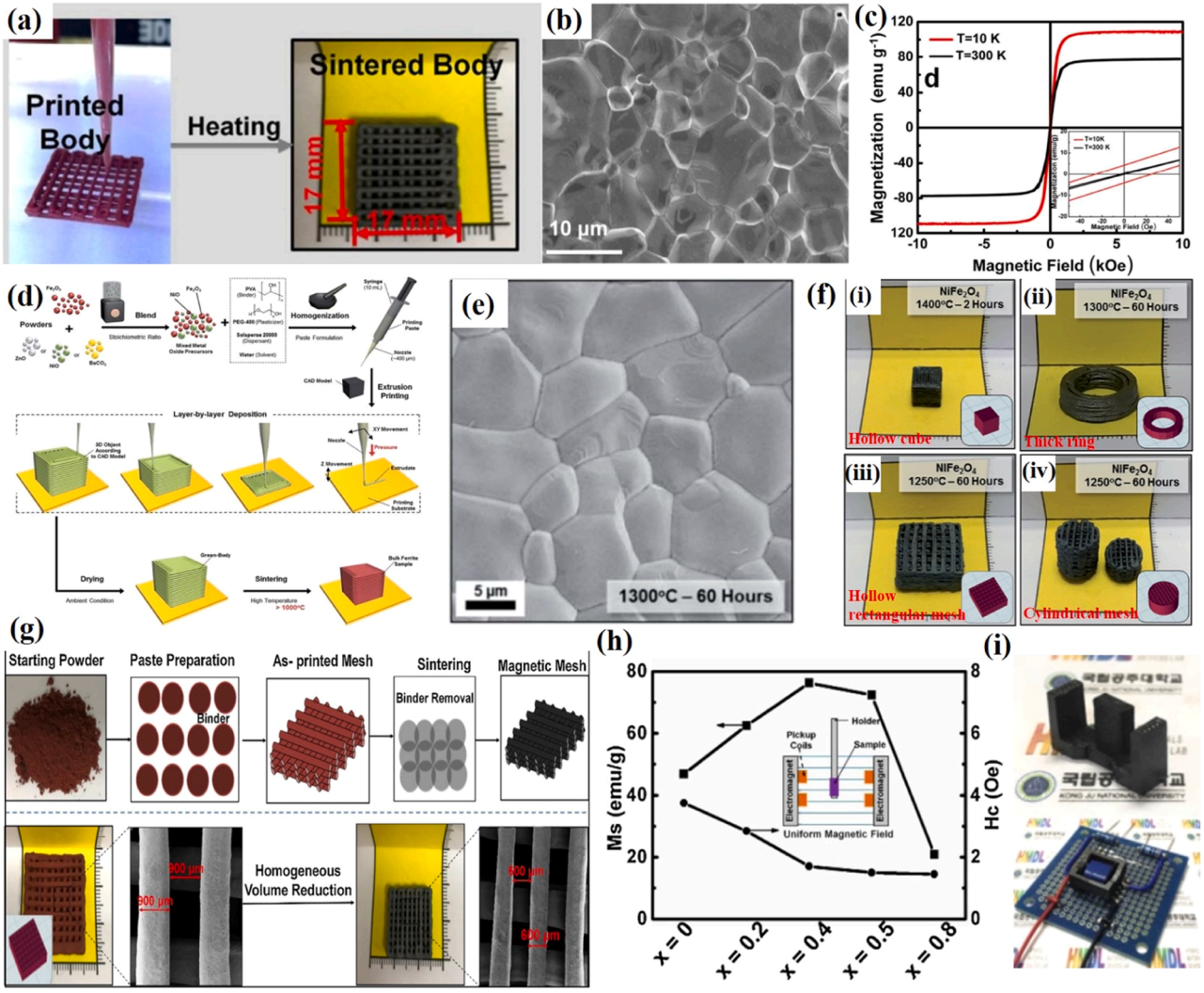

Fig. 18. DIW 3DP of soft magnetic ferrites (a) as-built and sintered NiZn ferrite samples [134]; (b) SEM diagram of NiZn ferrite [134]; (c) hysteresis loops at temperature of 10 K and 100 K [134]; (d) schematic diagram of FFF Ni ferrite [135]; € SEM diagram of Ni ferrite; (f) $NiFe_2O_4$ with complex structure [135]; (g) 3DP of NiZn ferrite with mesh architectures [136]; (h) M_s and H_c of 3DP NiZn ferrites with different Zn concentrations [136]; and (i) transformer with EFD core [137].

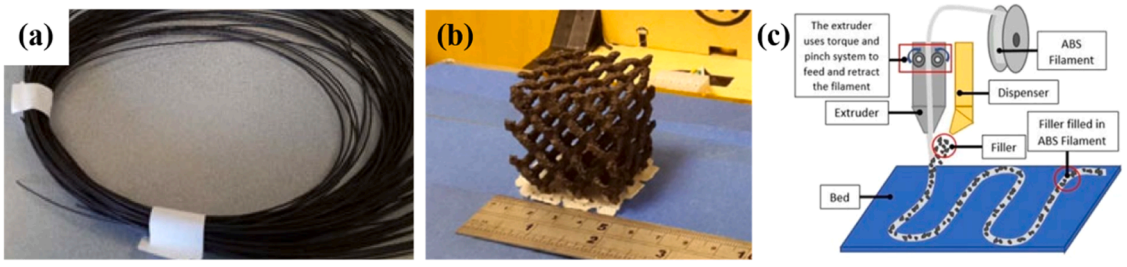

Fig. 19. (a) ABS/NZF composite filament [138]; (b) ABS/NZF with diamond lattice structure printed using FDM [138]; (c) schematic diagram of FDM 3DP of ABS/$ZnFe_2O_3$ [139].

[141]. The results show that the inductance of the inductor is almost constant at frequencies up to 1 MHz (Fig. 20b) [141]. Kim et al. studied IJP NiZn ferrite films for flexible electronic products [142–144]. Bissannagari [142] reported that IJP NiZn ferrite films annealed at 950 °C exhibit desirable soft magnetic properties [142], as shown in Fig. 20c. The resistance of the NZF layer embedded in PDMS remained approximately constant up to 5.5 % strain (Fig. 20d) [142]. In addition, the wireless power transmission (WPT) performance of the flexible NZF film was confirmed by the light-emitting diode (Fig. 20e) [142]. NiZn ferrites with trench structure are also fabricated by IJP [143], as shown in Fig. 20f. The conductive ink was printed into the 3D spiral trench structure to form a high-aspect-ratio 3D spiral inductor coil with a resistance of 26 Ω [143].

MZF is another soft magnetic ferrite commonly used in 3DP. Hrakova [145] reported the first fluxgate with IJP MZF cores, as shown in Fig. 21a. Although this fluxgate had a wide open-loop linear range of ±2 mT, its sensitivity was 10 mV/mT, which is considerably low [145]. In another study, MZFs were prepared by VP [146], as shown in Fig. 21b.

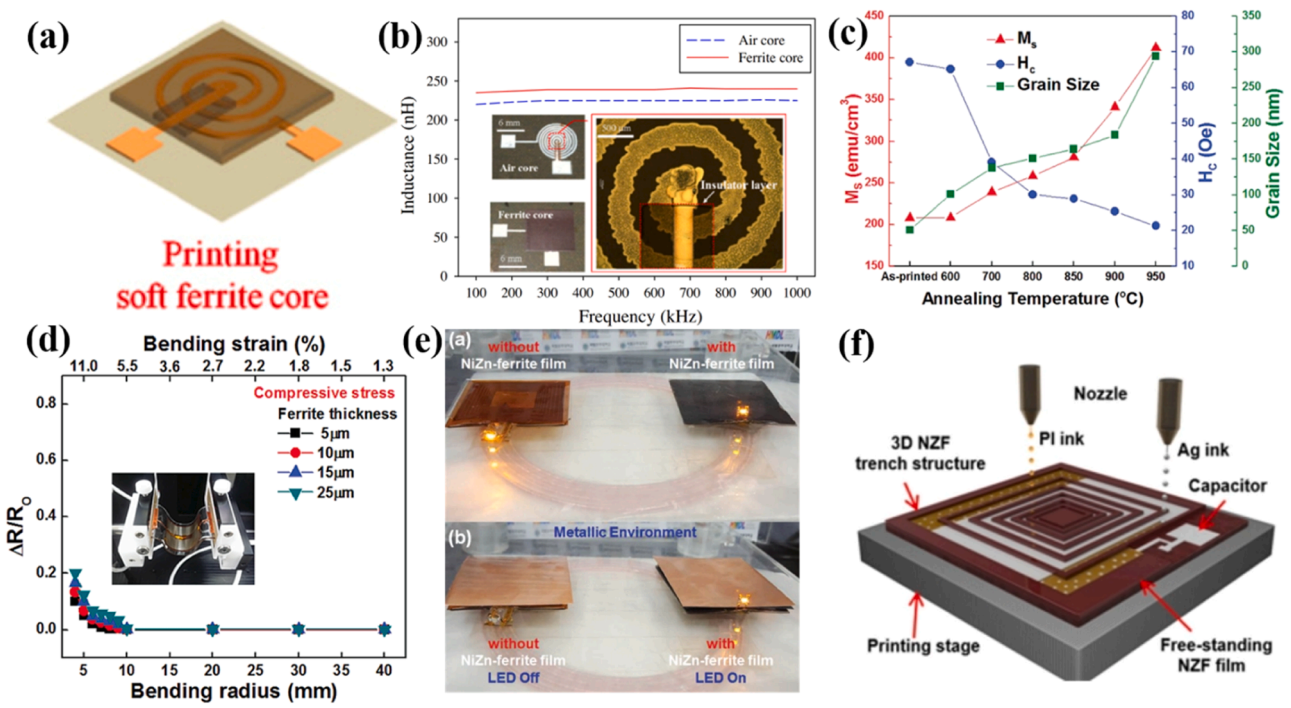

Fig. 20. (a) IJP Flat spiral inductors with soft ferrite films [141]; (b) inductance at high frequency [141]; (c) effect of annealing temperature on M_s and H_c of IJP ferrite films [142]; (d) normalized resistance of flexible ferrite films with different thickness [142]; (e) WPT performance testing [142]; and (f) high-aspect-ratio 3D spiral inductor coil [143].

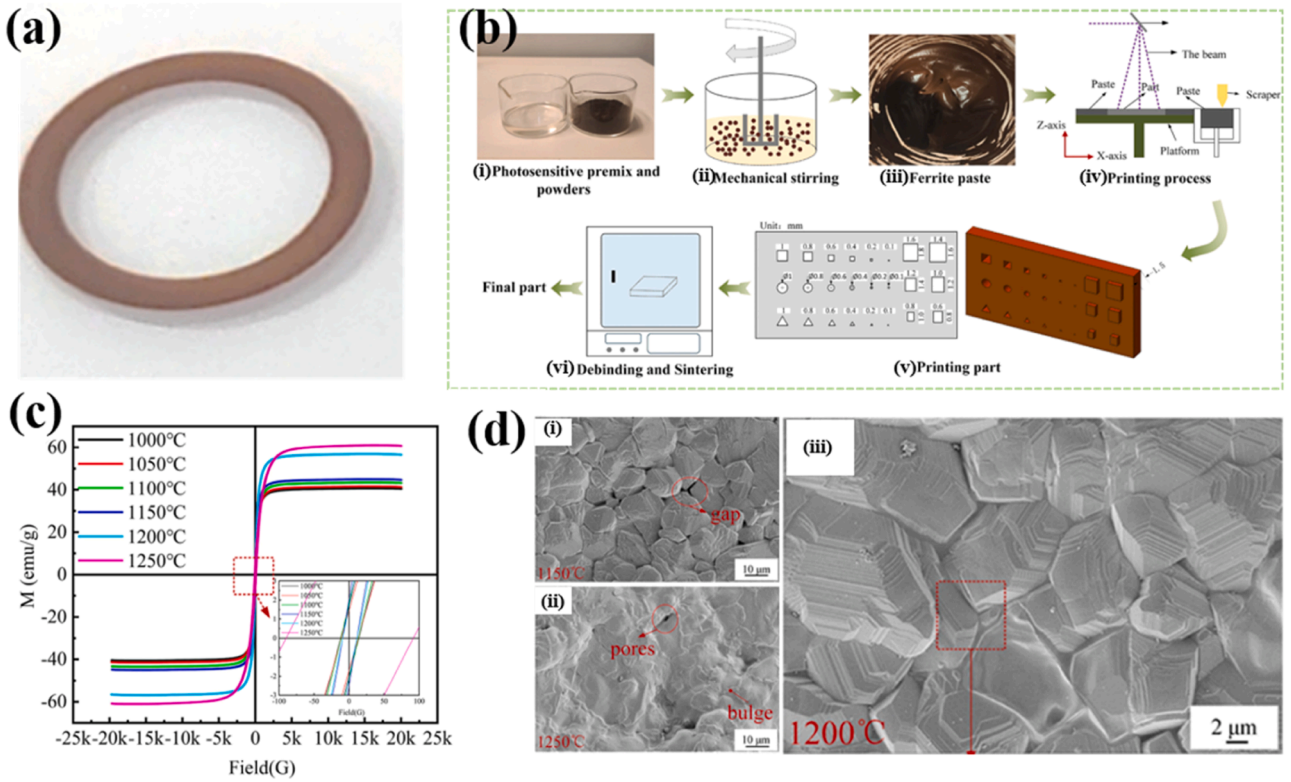

Fig. 21. (a) IJP Mn-Zn ferrite core [145]; (b) schematic diagram of SLA Mn-Zn ferrite [146]; (c) effect of sintering temperature on soft magnetic properties of SLA Mn-Zn ferrite [146]; and (d) SEM images of Mn-Zn ferrites sintered at different temperatures [146].

The optimal process parameters were reported to be: scanning space of 2 μm, scanning speed of 2500 mm/s, and laser power of 180 mW [146]. The density of the sintered parts reached 4.04 g/cm^3, and the M_s = 58.6 emu/g [146], as shown in Fig. 21c. This is mainly attributed to the compact microstructure of the sintered parts at 1200 °C (Fig. 21d).

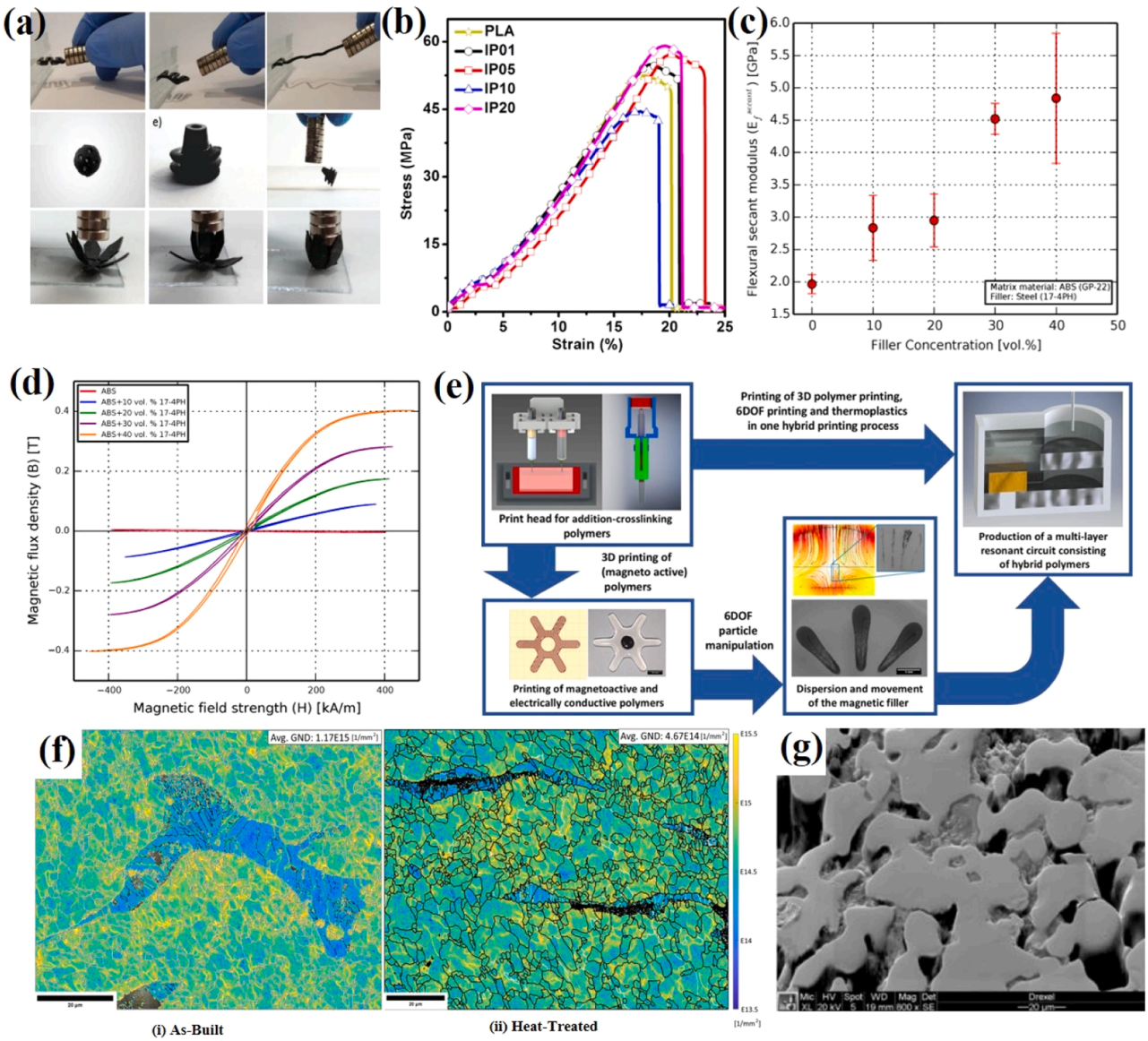

Fig. 22. (a) Magnetic responses of different SMCs [153]; (b) and (c) tensile properties of carbonyl iron particles filled thermoplastics composites [154]; (d) characterization of magnetic properties of 3DP ABS/stainless steel SMCs [155]; (e) schematic diagram of DIW MAP components [159]; (f) GND density of L-PBF iron/NiZnCu-ferrite SMCs [160]; and (g) SEM diagram of BJ SMCs [161].

2.2.4. Soft Magnetic Composites

Soft magnetic composites (SMCs) are formed by uniformly dispersing magnetic particles in non-magnetic materials [147,148]. Dispersing magnetic metal particles in non-conducting objects can reduce high-frequency eddy current loss and improve application frequency. SMCs can be manufactured by VP [149,150], FDM [122], DIW [151] and PBF [152]. Lantean et al. printed magnetically responsive polymer materials with tunable mechanical and magnetic properties using DLP and adjusted the magnetic response of the sample by changing the loading of Fe_3O_4 nanoparticles [153]. Different printed samples showed different stiffnesses and magnetic responses, as shown in Fig. 22a. Guan et al. used FDM to print carbonyl iron particles-filled thermoplastics composites [154]. The composites exhibited hysteresis loops of typical soft ferromagnets with relatively low remanence, coercive force, and hysteresis losses [154]. Moreover, no linear relationship was observed between magnetization and filler content. Compared with the traditional manufacturing method, FDM SMCs exhibited a significant increase in total elongation and a slight increase in tensile strength [154], as shown in Fig. 22(b and c). Polymer SMCs for 3DP were developed using ABS as the matrix and stainless steel powder as the filler [155]. The flexural secant modulus of the 3DP composite is twice that of pure ABS [155]. Magnetic characterization (Fig. 22d) of the composites showed that the samples with 40 vol% steel exhibited a magnetic retentivity of 15.6 mT and H_c of 54.16 Oe at an applied field of 485 kA/m [155].

Zhu [156] et al. developed a new composite ink system composed of PDMS and iron nanoparticles for DIW 3DP. The obtained sample showed a rapid response to a magnetic stimulus. Moreover, iron powder-based magnetorheological elastomers can be printed by DIW [157,158]. Sindersberger et al. tried to print magnetoactive polymer (MAP) components by DIW (Fig. 22e), which greatly expanded the application possibility of magnetoactive and electroactive polymer [159] and opened up new possibilities for manufacturing soft magnetic material components. Andrews [160] et al. prepared iron and NiZnCu-ferrite powder SMCs by L-PBF. EBSD (Fig. 22f) results show that the geometrically necessary dislocation(GND) density in the NiZnCu-ferrite region is lower than that in the iron phase [160]. Moreover, after heat treatment, the GND density in the iron phase decreases by 10 %, while the

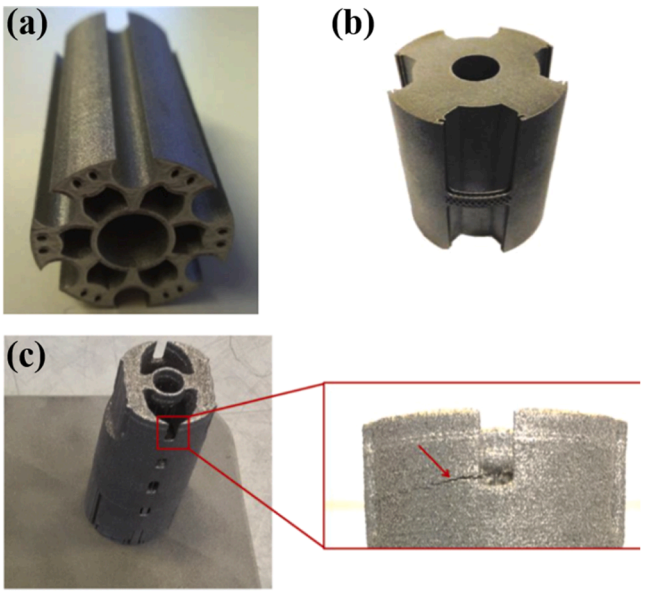

Fig. 23. Components built using SLM: (a) Fe–Co rotor core of switch reluctance machine [165], (b) novel skew rotor [166], and (c) Fe-6.9Si rotor core [167].

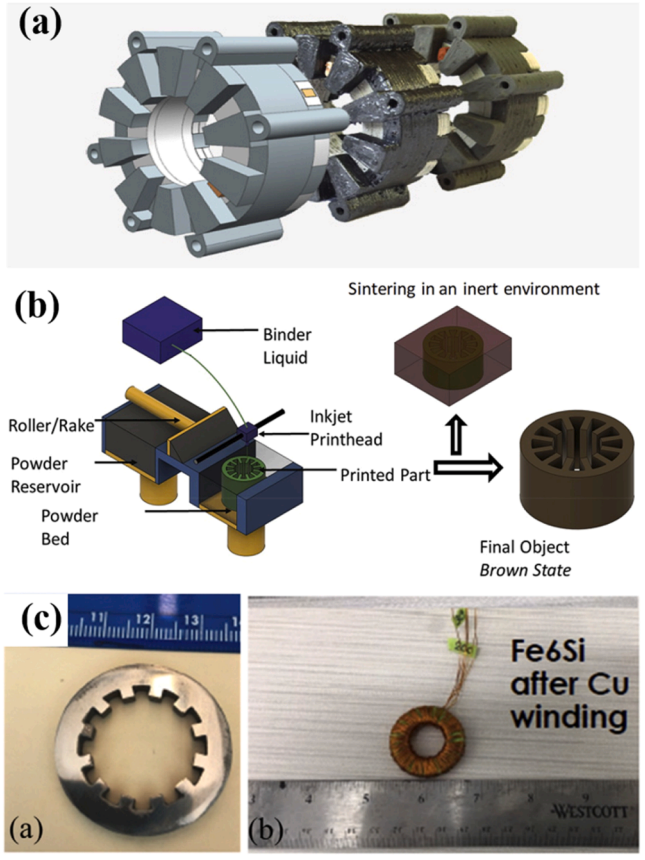

Fig. 24. (a) FDM switched reluctance machine [168]; (b) schematic diagram of BJ shaped stator core [168]; and (c) BJ Fe-6.5Si stator ring [73].

ferrite phase increases by 4 %. In addition, BJ can also be used to print SMCs. For example, [161] reported that iron powder samples coated with phosphorus were first sprayed via BJ, then solidified at 816 °C to burn off the binder, and finally, annealed at 1260 °C [161]. The porosity of the final sample is approximately 35 %, as shown in Fig. 22g.

3. Applications of 3DP SMMs

The ultimate goal of 3DP of SMMs is their application. 3DP soft magnetic devices provide remote controllability and fast response. Currently, the application scenarios of 3DP SMMs mainly include electronic components such as transformers, inductors, and micro-actuators; soft robots; electromagnetic microwave absorption; and magnetic separation. This part discusses the applications of 3DP soft magnetic in different fields.

3.1. Electronic components

SMMs are ideal for power electronics applications. In power electronics systems, 3DP allows for the formation of complex-shaped parts to achieve the desired functionality. Motors, transformers, inductors, micro-actuators, and energy harvesters are key components of most electronic products.

Some scholars have attempted to use the SLM to print the rotor core of small motors [162–164]. The commonly used materials for SLM manufacturing of motor cores are Fe-Si and Fe-Co alloys. Fig. 23 shows some examples of SLM rotor cores. In [165], a Fe-Co rotor core with a complex structure was fabricated using SLM, as shown in Fig. 23a. The saturation magnetization and mechanical properties of the core are comparable to those of the standard Fe-Co alloys [165]. A novel switched reluctance machines(SRM) rotor structure is presented to decrease the torque ripple and windage loss [166], as shown in Fig. 23b. Nevertheless, there are also some quality concerns in the magnetic parts prepared by SLM, such as the occurrence of cracks during the printing process, as shown in Fig. 23c. Therefore, further optimization of 3DP is needed in future.

In [168], 3DP motors were studied. FDM was used to print a switched reluctance machine, which was then sintered, as shown in Fig. 24a. National Aeronautics and Space Administration(NASA) is committed to studying FDM of complex geometry stators to achieve high power density aviation motors [169]. The stator core can also be manufactured using BJ (Fig. 24b). Cramer et al. reported a new method to fabricate a soft magnetic Fe-6.5Si stator ring via BJ [73]. After annealing, the ring core was wound with multistrand Litz wire, as shown in Fig. 24c. As-annealed samples exhibited a low core loss, high resistivity, and good magnetic permeability [73].

The transformer core can also be realized using 3DP. For example, Plotkowski [92] et al. fabricated Fe-6Si desktop transformer cores by SLM and conducted magnetic tests (Fig. 25a). These cores exhibited low power losses across all measured polarizations [92], as shown in Fig. 25b. Bollig [122] et al. successfully manufactured a complex transformer core using FDM and pointed out that high filling factor and high concentration of magnetic particles are needed to ensure the high core performance. Lazarus [170] et al. proposed a method for fluid 3DP magnetic devices. Liquid metals are combined with a magnetic core to create complete magnetic core 3DP inductors [170]. This method has been used to print magnetic core inductor and magnetic transformer, as shown in Fig. 25(c,d), respectively. DIW has been successfully applied for manufacturing magnetic transformers [137]. Fig. 25e shows the final assembly of the DIW-EFD transformer core and the EFD transformer. Notably, 3DP of a toroid inductor has been demonstrated (Fig. 25f) [171]. The inductance of this inductor in the frequency range of 1 kHz to 3 MHz is 167 nH-143 nH, and the DC resistance of its winding is 0.1 Ω.

3.2. Magnetic-drive robot

SMMs are widely used in robots due to their excellent mechanical response to EMF. The magnetic-drive robot can be realized by mixing magnetic particles into the polymer matrix or coating magnetic particles on the polymer surface. However, robots with simple structures can only achieve a single function, and it is difficult to achieve a combination of multiple functions. 3DP can manufacture multi-material and complex

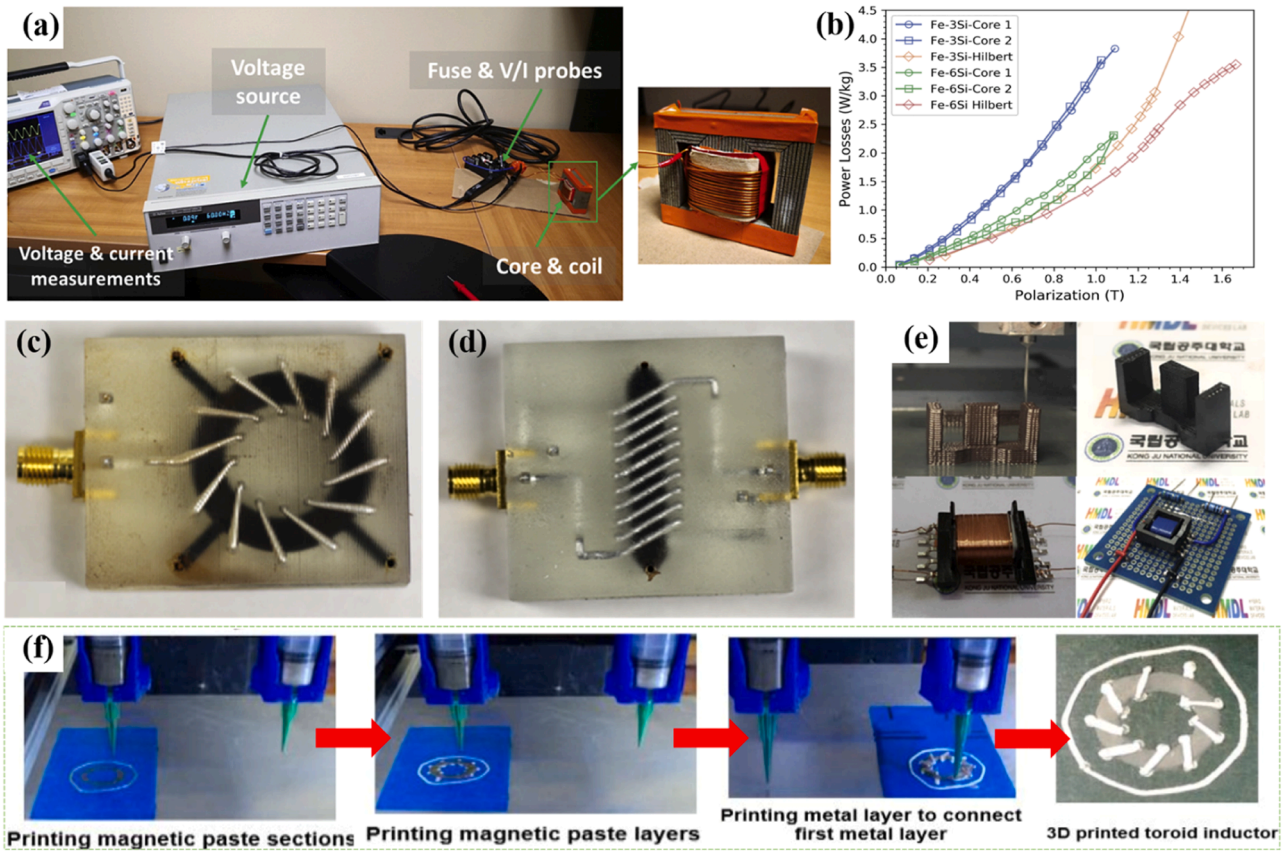

Fig. 25. (a) SLM Fe-6Si desktop transformer [92]; (b) transformer performance test [92]; (c) annular core inductor [170]; (d) magnetic transformer [170]; (e) DIW-EFD transformer [137]; and (f) DIW ring inductor [171].

structural parts, thus enabling the design and manufacturing of magnetic-drive robots with integrated structure and function.

Inspired by the inchworm, Joyee et al. proposed a multi-material, multi-functional soft monomer robot with two-axis motion ability [172,173]. The robot is made of magnetic particle polymer by 3DP (Fig. 26a). The presence of magnetic particles allows the robot to perform various motion functions such as crawling and rotating [172]. Moreover, this robot can perform controlled virtual drug delivery, as shown in Fig. 26b. Magnetic photosensitive resin for DLP 3DP has been developed with Fe_3O_4 nanoparticles as filler [174]. A magnetically-driven flexible gripper robot with a complex structure was printed using DLP (Fig. 26c) [174]. Its sensitive magnetic response easily grasped the target object. In addition, Zhu et al. developed an SLA-based rapid 3DP technology called microscale continuous optical printing to achieve functional artificial microfish with different bionic structures and movement capabilities. Magnetic Fe_3O_4 nanoparticles with a concentration of 5 mg/mL were embedded in the matrix material of the fish head [175], thus achieving a directional movement under magnetic drive, as shown in Fig. 26d.

Soft actuators based on ferromagnetic fluids have been manufactured using 3DP without embedding magnetic particles into polymer matrices [176]. In particular, a chiral actuator was fabricated and demonstrated for complex motions, as shown in Fig. 26e. The ferromagnetic fluid was moved to different positions to affect the actuator response. DIW has also been used for the printing of magnetic-drive robots. Roh et al. demonstrated a new class of magneto-capillary 3DP soft actuators(Fig. 26f) [177]. Under the action of EMF, these actuators captured objects floating on water. They also pointed out that this new actuator realized by 3DP can be used for manufacturing various types of soft robots with numerous potential applications [177]. Some bionic robots inspired by mollusks have been developed using 3DP. These include starfish-type robots (Fig. 26g and h) [178,179] and worm-type robots (Fig. 26i) [180].

3.3. Other applications

In addition to motors, transformers, inductors, and magnetically driven robots, 3DP SMMs are used in devices for electromagnetic microwave absorption, magnetic separation, and drug delivery. Xiong et al. printed Fe-Si-Al/PP material with a high loading fraction using FDM, as shown in Fig. 27a [181]. They reported that, for magnetic powder content of 94 wt%, the effective permeability was 16.5 in the frequency range of 1 kHz–10 MHz. Furthermore, the core loss was 11.22 W/kg when the magnetic excitation level was set at H_m =1000 A/m, and the frequency was 65 kHz; moreover, the maximum quality factor was 79.5, and the M_s was 123 emu/g.Ye et al. studied the electromagnetic wave absorption properties of graphene/FeSiAl/PLC composites prepared by FDM [182]. The composites showed improved electric magnetic field (EMW) absorption performance when the graphene content was increased. Specifically, the 4 wt% GR/FeSiAl/PLA composites exhibited a minimum reflection loss (RL) value of −46.43 dB and an effective absorbing bandwidth (RL≤−10 dB) of 4.80 GHz (13.2–18 GHz) at 2.2 mm [182], as shown in Fig. 27b. The minimum RL value is −31.53 dB and the maximum effective absorbing bandwidth(EAB) is 6.40 GHz (11.6–18 GHz) at 2.2 mm when the graphene content is increased to 5 wt% [182], as shown in Fig. 27c. Based on [182], aiming to improve the bandwidth of the absorber, a novel FeSiAl/PLA composites pyramid(Fig. 27d) was fabricated by FDM [183]. The results showed that the minimum RL of the pyramid absorber was −37.2 dB, and the maximum EAB was 11.3 GHz. Another study analyzed the microwave absorption properties of 3DP $Li_{0.44}Zn_{0.2}Fe_{2.36}O_4$/PLC composites in terms of fused deposition modeling [184]. When the content of

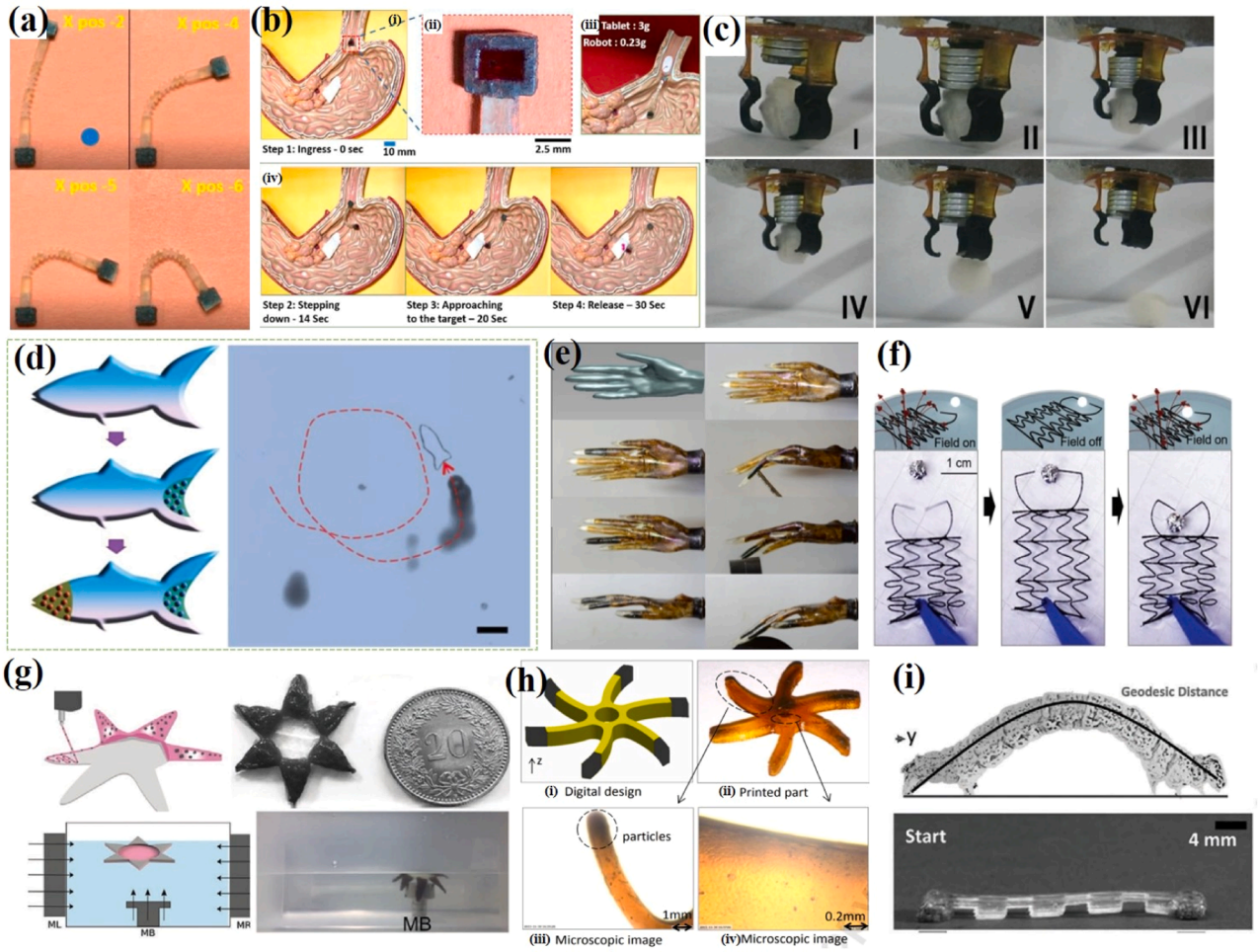

Fig. 26. (a)3DP of a soft robot [172]; (b) magnetic-drive robot that transports virtual drugs [172]; (c) DLP magnetic-drive flexible gripper [174]; (d) μCOP functional artificial microfish [175]; (e) chiral actuator of ferromagnetic fluid controls [176]; (f) DIW magnetic-drive soft actuators [177]; (g) and (h) Starfish-type magnetically driven robots [178,179]; and (i) worm-type magnetically driven soft robot [180].

$Li_{0.44}Zn_{0.2}Fe_{2.36}O_4$ was 20 %, the composite material showed good absorbing properties, as shown in Fig. 27e. The maximum RL value was −32.4 dB at 3.8 GHz and −31.8 dB at 16.1 GHz with the thickness of 6 mm, and the maximum EAB corresponding to RL (≤−10 dB) was 2.1 GHz (3.1–5.2 GHz) and 1.8 GHz (14–15.8 GHz) [184]. Therefore, microwave-absorbing materials prepared by 3DP may become a development trend in the future. Magnetic separation enables the use of an external magnetic field to collect materials from water or other liquids. Wei et al. prepared a network structure with high magnetic susceptibility using 3DP to achieve magnetic separation effectively under a low magnetic field, as shown in Fig. 27f [134–136]. This material with complex structures is a promising candidate for water purification. 3DP SMMs can be applied to drug delivery using a working principle similar to that of magnetic-drive robots. Integration of magnetic particles could increase the magnetism of the original materials [185] and realize directional movement under the action of an external magnetic field. A magnetic tubular microstructure was fabricated by 3DP for targeted drug delivery [186]. Under guiding and release mechanisms, this micromotor can deliver a drug to targeted tumor cells, as demonstrated in Fig. 27g.

4. Challenges and opportunities associated with 3DP of SMMs

Although the research on 3DP of SMMs has progressed significantly, many challenges must be addressed before realizing more practical applications:

Most of the effort must be focused on process optimization to obtain superior soft magnetic properties of 3DP SMMs. In practical applications, soft magnetic performance is just one of the required properties. For example, it is also important to achieve high strength and low power loss in electronic equipment. The influence of multiple variables on properties should be carefully studied in the 3DP of SMMs. In addition, the underlying mechanism among the raw materials, processes, microstructures, and properties of 3DP SMMs must be clarified to form a principle rule that can guide future work.

The concentration of magnetic filler is an important factor affecting the performance of 3DP SMMs. The concentrations of magnetic fillers in slurry or ink are relatively low in current research. This approach is necessary to meet the needs of 3DP methods such as FDM, DIW, and IJP and ensure a smooth printing process. The base matrix with high volume fraction magnetic filler limits the choices of applicable 3DP technologies. Therefore, fabricating a high-concentration magnetic base matrix by 3DP is another problem for 3DP of SMMs. This also poses a challenge to the development of new 3DP technologies. In addition, fewer SMMs are available for 3DP. To make the application of 3DP SMMs more extensive, exploring other SMMs for 3DP is necessary.

High-precision fabrication of multi-material components has become a trend. 3DP SMMs combined with other functional materials will allow engineers to design more complex devices and diverse functions. Therefore, the high-precision manufacturing of multi-material, multi-functional, and complex structure magnetic parts is another challenge in this field.

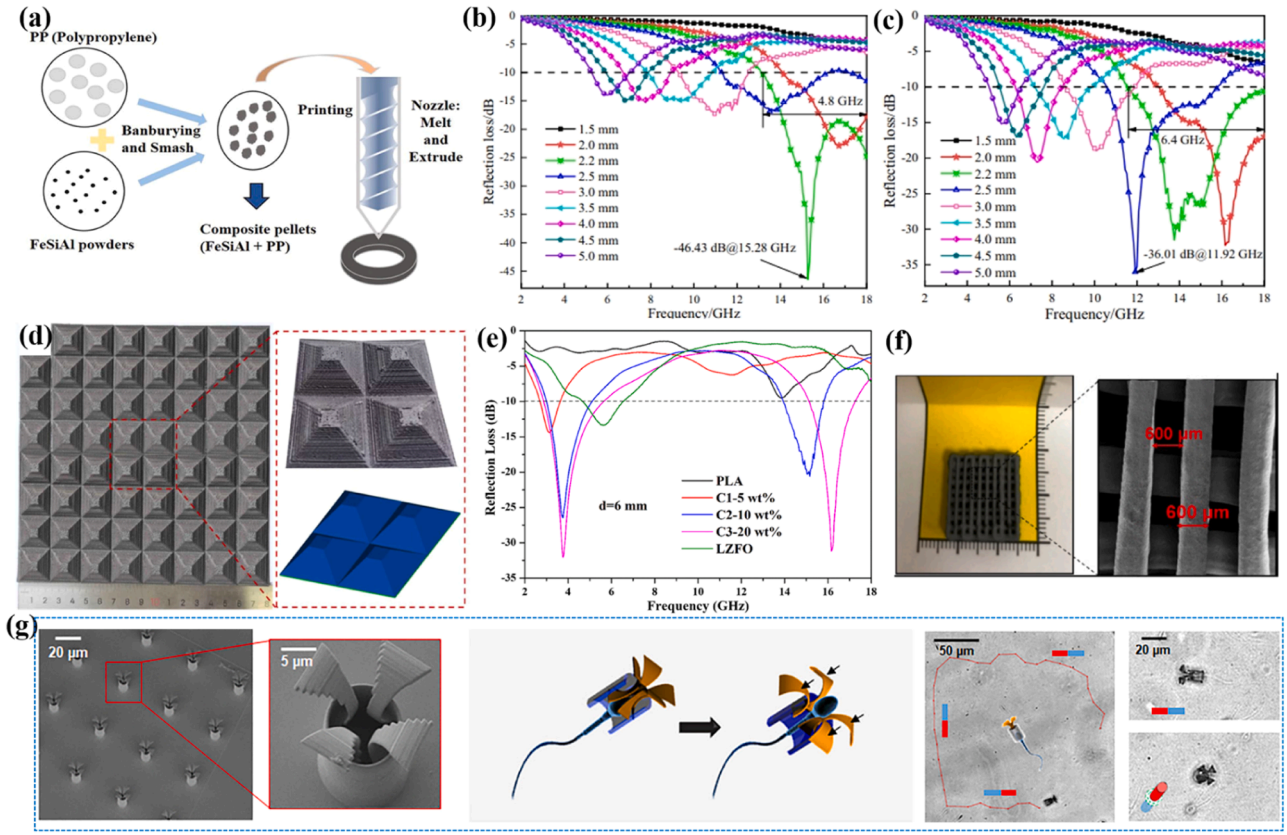

Fig. 27. (a) FDM printed Fe-Si-Al/PP material [181]; (b) the RL value of 4 wt%GR/FeSiAl/PLA composites; (c) the RL value of 5 wt%GR/FeSiAl/PLA composites [182]; (d) FeSiAl/PLA composites pyramid [183]; (e) absorption properties of 3DP $Li_{0.44}Zn_{0.2}Fe_{2.36}O_4$/PLC composites [184]; (f) 3DP network structure for magnetic separation [136]; and (f) 3DP magnetic tubular microstructure for targeted drug delivery [185].

Compared with other preparation technologies, 3DP offers unique advantages for SMMs, such as high design flexibility, rapid prototyping, efficient resource utilization, complex structure realization and customized production. Overall, 3DP offers a versatile and efficient approach to manufacturing SMMs, unlocking new possibilities for design freedom, customization, and innovation. Research on 3DP of SMMs is still in its preliminary stages, and compared to other preparation technologies and/or materials, there are still many apparent drawbacks, such as limited material selection, unstable surface quality, high cost and material performance instability. Despite these drawbacks, it is expected that the disadvantages of 3DP for SMMs will gradually be alleviated with the continuous development and improvement of the technology.

5. Summary and outlook

Studies on the preparation of different SMMs using different 3DP techniques have been reviewed in this paper. Reports from various research groups indicate that 3DP is a feasible manufacturing process for developing soft magnetic components. 3DP can produce near-net-shaped magnetic parts with improved magnetism and reduced processing costs. Many SMMs, such as Fe-Si and Fe-Ni alloys and soft magnetic ferrites and composites, have been successfully fabricated using 3DP technologies such as SLM, FDM, DIW, BJ, and IJP. The performance of 3DP soft magnetic components depends on the type of raw materials, volume fraction of magnetic particles, process parameters, scan pattern and geometry, and post-treatment. These key factors influence product performance by regulating the microstructure. In addition, we focused on the application scenarios of 3DP SMMs, including electronic components such as transformers, motors, inductors, sensors, magnetic-drive robots, flexible electronics, and devices used for microwave absorption and magnetic separation. 3DP promotes the manufacturing of integrated and flexible electronic devices. Advanced 3DP technologies have been used to build well-designed magnetically-driven robots with high driving efficiency. 3DP of SMMs is currently in its initial stage and involves multiple disciplines such as electromagnetism, physics, material science, and mechanics. A wide space exists for researchers to develop 3DP technology for advanced SMMs and devices.

CRediT authorship contribution statement

Fenghui Wang: Writing – original draft. **Xiaopei Zhu:** Data curation. **Jing Zhang:** Project administration. **Heguang Liu:** Methodology. **Na Tian:** Supervision. **Caiyin You:** Writing – review & editing.

Declaration of Competing Interest

We hereby confirm that this manuscript is our original work and has not been published nor has it been submitted simultaneously elsewhere. We further confirm that all authors have checked the manuscript and have agreed to the submission.

Data availability

Data will be made available on request.

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